

On Equivalence of the Butterfly Structure

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Abstract

The butterfly structures were introduced by Perrin et al. as a generalization of the Dillon's APN permutation operated in 6 bits. It was proved by several authors independently that such butterflies are differentially 4-uniform and have best known nonlinearity among functions over $\mathbb{F}_{2^{2k}}$. In [21, 22] authors provided sufficient coefficient conditions such that the closed butterfly is a permutation and they further prove that such functions have boomerang uniformity 4 and are linear equivalent to Gold functions. In this paper, we adopt techniques from [10] and [13] to classify closed butterfly structures satisfying more general conditions in terms of EA-equivalence and CCZ-equivalence.

Keywords: Butterfly structure, bijective functions, linear equivalence, EA-equivalence, CCZ-equivalence

MSC Classification: 11T06, 11T71

1 Introduction

1.1 Background

Let $q = 2^k$ for some positive integer k and $\mathbb{F}_q$ denote the finite field with q elements. Given a bivariate function $R: \mathbb{F}_q \times \mathbb{F}_q \rightarrow \mathbb{F}_q$ such that $R_y: x \mapsto R(x, y)$ permutes $\mathbb{F}_q$ for all $y \in \mathbb{F}_q$, the closed butterfly with respect to R is given by

$$\mathbf{V}_R(x, y) := (R(x, y), R(y, x)). \quad (1)$$

The open butterfly with respect to R is given by

$$\mathbf{H}_R(x, y) := (R(y, R_y^{-1}(x)), R_y^{-1}(x)),$$

where R_y^{-1} is the (compositional) inverse function of R_y .

These structures were first introduced by Perrin et al. [24] with R in the form $R(x, y) = (x + \alpha y)^e + y^e$ where $\alpha \in \mathbb{F}_q$ and e is a positive integer with $\gcd(e, q-1) = 1$. It was motivated by the discovery of the first known APN permutation over $\mathbb{F}_{2^6}$ by Dillon, which is linear equivalent to the open butterfly $\mathbf{H}_R$ over $\mathbb{F}_{2^3} \times \mathbb{F}_{2^3}$ with respect to this R with α being a root of the polynomial $X^3 + X + 1 \in \mathbb{F}_2[X]$ and $e = 6$ (or 3). The butterflies $\mathbf{V}_R$ and $\mathbf{H}_R$ for general R were proposed and studied by Canteaut et al. [6].

It is known that the open butterfly $\mathbf{H}_R$ is always an involution (and hence a permutation) over $\mathbb{F}_q \times \mathbb{F}_q$, and is CCZ-equivalent to the corresponding closed butterfly $\mathbf{V}_R$. Hence both butterflies for the same R share the same differential spectra (and hence differential uniformity) and Walsh spectra (and hence nonlinearity). Several later works ([6, 12]) used this fact to generalize the result in [24], and finally in [23]

authors considered the function

$$R(x, y) = R_{\alpha, \beta}^{Q+1}(x, y) := (x + \alpha y)^{Q+1} + \beta y^{Q+1}$$

for $\alpha, \beta \in \mathbb{F}_q$ and $Q = 2^\ell$ for some positive integers ℓ , such that k is odd and $\gcd(k, \ell) = 1$. They proved that as long as $\alpha, \beta \in \mathbb{F}_q^*$ with $\beta \neq (\alpha + 1)^{Q+1}$, the closed butterfly $V_{\alpha, \beta}^{Q+1}$ with respect to $R_{\alpha, \beta}^{Q+1}$ has differential uniformity at most 4 and nonlinearity $2^{2k+1} - 2^k$. The latter quantity is the best nonlinearity known for functions over $\mathbb{F}_{2^{2k}}$. The corresponding open butterfly $H_{\alpha, \beta}^{Q+1}$ is a permutation with best known nonlinearity, low differential uniformity as well as large algebraic degree. In [7], Canteaut et al. further showed that there are no new APN butterflies in this construction except for the 6-bit butterflies that are CCZ-equivalent to Dillon's permutation.

While the open butterfly $H_{\alpha, \beta}^{Q+1}$ is always a permutation from the construction, the corresponding closed butterfly $V_{\alpha, \beta}^{Q+1}$ may not be so. In [22], Li et al. provided sufficient conditions such that $V_{\alpha, \beta}^{Q+1}$ is a permutation, and furthermore they proved that such closed butterflies have boomerang uniformity 4, another interesting cryptographical quantity recently discovered to analyze boomerang attack. As a supplementary note to this, in [21] authors derived a simpler but equivalent condition for this, and showed that such permutations are actually linear equivalent to a Gold function. More precisely, these results can be summarized as follows:

Theorem 1 *Let $q = 2^k$ and $Q = 2^\ell$ for positive integers k, ℓ such that k is odd and $\gcd(k, \ell) = 1$. For $\alpha, \beta \in \mathbb{F}_q$, define $R_{\alpha, \beta}^{Q+1}(x, y) = (x + \alpha y)^{Q+1} + \beta y^{Q+1} \in \mathbb{F}_q[x, y]$. Let $V_{\alpha, \beta}^{Q+1}$ be the closed butterfly with respect to $R_{\alpha, \beta}^{Q+1}$ as defined in (1), and define the set*

$$\Gamma := \{(\alpha, \beta) \in \mathbb{F}_q^* \times \mathbb{F}_q^* : \varphi_2^Q = \varphi_1 \varphi_3^{Q-1}, \varphi_3 \neq 0\},$$

where φ_i 's are defined by

$$\begin{cases} \varphi_1 &= (\alpha + 1)^{2Q+2} + \alpha^{2Q+1} + \alpha^Q + \alpha\beta + \beta^2 \\ \varphi_2 &= (\alpha + 1)^{2Q+2} + \alpha^{Q+2} + \alpha + \alpha^Q\beta + \beta^2 \\ \varphi_3 &= (\alpha + 1)^{2Q+2} + \beta^2. \end{cases}$$

Then the following holds.

1. ([22, Theorem 1]) If $(\alpha, \beta) \in \Gamma$, then $V_{\alpha, \beta}^{Q+1}$ is a permutation on $\mathbb{F}_q \times \mathbb{F}_q$ with boomerang uniformity 4;
2. ([21, Theorem 2]) If ℓ is odd,
 - (a) $(\alpha, \beta) \in \Gamma$ if and only if $\alpha, \beta \in \mathbb{F}_q^*$ satisfy $\alpha^2 + \alpha\tilde{\beta} + \tilde{\beta}^2 = 1$, where $\tilde{\beta}$ is the unique $(Q+1)$ -th root of β in $\mathbb{F}_q^*$;
 - (b) if $(\alpha, \beta) \in \Gamma$, then $V_{\alpha, \beta}^{Q+1}$ is linear equivalent to the Gold function $X^{2^{k-\ell}+1}$ over $\mathbb{F}_{q^2}$.

It is remarked that from the proof in [21], in fact Statement 2 (a) of Theorem 1 also holds for even ℓ . It was conjectured that $(\alpha, \beta) \in \Gamma$ is also necessary for $V_{\alpha, \beta}^{Q+1}$ to be a permutation, and later in [10, Corollary 9.5] Ding and Zieve studied the geometric properties of related rational functions to confirm this conjecture affirmatively, using the equivalent condition in Statement 2 (a) of Theorem 1.

From all these, it is a natural question to ask what happens to $V_{\alpha, \beta}^{Q+1}$ when $(\alpha, \beta) \notin \Gamma$, say what simple function it is equivalent to. In [21, Remark 2 (3)], authors made experiments to show that when $k = 3, 5$ and $(\alpha, \beta) \notin \Gamma$, $V_{\alpha, \beta}^{Q+1}$ is CCZ-inequivalent to Gold functions. In this paper we resolve and extend this observation partially to general odd k .

1.2 Statement of main results

The main result of our paper is now summarized as follows. For any positive integer n , recall that a *2-cyclotomic coset modulo n* is a subset of $\mathbb{Z}/n\mathbb{Z}$ of the form

$$C_s = \{s, 2s, 2^2s, \dots, 2^{r-1}s\} \pmod{n}.$$

Here we write each element of C_s as an integer between 0 and $n - 1$. Moreover, the *representative* of C_s is the smallest integer in the coset.

Theorem 2 *Let $q = 2^k$ and $Q = 2^\ell$ for positive integers k, ℓ such that k is odd and $\gcd(k, \ell) = 1$. For $\alpha, \beta \in \mathbb{F}_q^*$, define $R_{\alpha, \beta}^{Q+1}(x, y) = (x + \alpha y)^{Q+1} + \beta y^{Q+1} \in \mathbb{F}_q[x, y]$. Let $V_{\alpha, \beta}^{Q+1}$ be the closed butterfly with respect to $R_{\alpha, \beta}^{Q+1}$ as defined in (1). Let θ be a fixed primitive element of $\mathbb{F}_q^*$, and $\mathcal{C}$ denote the set of all nonzero 2-cyclotomic coset representatives modulo $(q - 1)$.*

Define the parameters

$$\Theta_1 := (\alpha + 1)^{2Q+2} + \beta^2$$

$$\Theta_2 := \alpha^{2Q+1} + \alpha + \alpha^Q \beta$$

$$\Theta_3 := \alpha^{Q+2} + \alpha^Q + \alpha\beta.$$

Assume $\Theta_1 \neq 0$.

1. *In the following, $\sim_L$ means “is linear equivalent to”.*

(a) *$V_{\alpha, \beta}^{Q+1}$ induces a permutation on $\mathbb{F}_q \times \mathbb{F}_q$ if and only if $\Theta_2^Q = \Theta_1^{Q-1} \Theta_3$. In this*

$$\text{case, } V_{\alpha, \beta}^{Q+1} \sim_L \mathcal{G}(X) := \begin{cases} X^{Q+q} \in \mathbb{F}_{q^2}[X] & (\ell \text{ odd}) \\ X^{Q+1} \in \mathbb{F}_{q^2}[X] & (\ell \text{ even}) \end{cases};$$

(b) *Otherwise,*

- *if $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 0$, then $V_{\alpha, \beta}^{Q+1} \sim_L F_i(x, y) := (x^{Q+1} + x^Q y, xy^Q + \theta^i y^{Q+1}) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y]$ for some $i \in \mathcal{C}$;*
- *if $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 1$, then $V_{\alpha, \beta}^{Q+1} \sim_L f_i(X) := X^{Q+q} + \theta^i X^{Q+1} \in \mathbb{F}_{q^2}[X]$ for some $i \in \mathcal{C}$.*

2. Let $\mathcal{P} = \{\mathcal{G}(X)\} \cup \{F_i(x, y) : i \in \mathcal{C}\} \cup \{f_i(X) : i \in \mathcal{C}\}$.

- (a) if $k \geq 3$, then the functions in $\mathcal{P}$ are pairwise EA-inequivalent. Hence the number of EA-equivalence classes of $\mathcal{V}_{\alpha, \beta}^{Q+1}$ with $\Theta_1 \neq 0$ is at most $1 + 2N$, where N is the number of elements of $\mathcal{C}$;
- (b) let $k \geq 3$, and r be a primitive prime divisor of $2^k - 1$, that is, a prime number such that $r \mid (2^k - 1)$ and $r \nmid (2^i - 1)$ for $1 \leq i \leq k - 1$, and R be the largest power of r dividing $q - 1$. If $(Q + 1)^k \not\equiv 1 \pmod{R}$, then the functions in $\mathcal{P}$ are pairwise CCZ-inequivalent. Hence the number of CCZ-equivalence classes of $\mathcal{V}_{\alpha, \beta}^{Q+1}$ with $\Theta_1 \neq 0$ is at most $1 + 2N$.

1.3 Discussion of main results

Comparing with the notations in Theorem 1, we see that $\Theta_1 = \varphi_3$, $\Theta_2 = \varphi_2 + \varphi_3$ and $\Theta_3 = \varphi_1 + \varphi_3$. Hence the conditions $\Theta_1 \neq 0$ and $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$ are equivalent to $(\alpha, \beta) \in \Gamma$ in Theorem 1. Moreover, when ℓ is odd, $\mathcal{G}(X) = X^{Q+q} \sim X^{qQ+1} = X^{2^{k-\ell}+1} \in \mathbb{F}_{q^2}[X]$. Therefore Theorem 2 covers all results in Theorem 1 as well as [10, Corollary 9.5].

It is well-known by Zsigmondy's Theorem (see [17, Chapter IX, Theorem 8.3]) that there exists a primitive prime divisor of $2^k - 1$ except when $k \in \{1, 6\}$, so that Statement 2(b) makes sense in our setting. Moreover, the conditions imply that k is the order of 2 modulo R . Since R is an odd prime power, $(Q + 1)^k \not\equiv 1 \pmod{R}$ is equivalent to saying that $Q + 1$ is not in the multiplicative group generated by 2 (mod R), or equivalently, there is no integer m such that $2^m \equiv Q + 1 \pmod{R}$. If k is such that $2^k - 1$ is already prime (so $R = r = 2^k - 1$; in this case the number $2^k - 1$ is called a Mersenne prime, and k is necessarily itself prime as well; the first few odd k 's satisfying this condition are $k = 3, 5, 7, 13, 17, 19, 31$), then it is clear that $2^m \not\equiv Q + 1 \pmod{R}$ for any integer m when $k \nmid \ell$. Hence under this case we know that the functions in $\mathcal{P}$ are pairwise CCZ-inequivalent. In fact by Magma experiment,

we find that for all odd k between 3 and 39, there exists a primitive prime divisor r of $q - 1$ such that the condition $(Q + 1)^k \not\equiv 1 \pmod{R}$ holds for all ℓ not divisible by k . Then Statement 2(b) implies that the functions in $\mathcal{P}$ are pairwise CCZ-inequivalent for all odd k between 3 and 39.

When $k = 3, 5$ (whence $q - 1 = 7, 31$ are both prime), N is 2 and 6 respectively, so $1 + 2N$ is 5 and 13 respectively. Magma experiments illustrate that the number of CCZ-equivalence classes of $V_{\alpha, \beta}^{Q+1}$ with $\Theta_1 \neq 0$ is exactly 5 and 13 respectively in these two cases (see [23, Section 4.4, Table 1] for $k = 5$). Hence the bound in Statement 2(b) is tight in these two cases. This means that for each function $P \in \mathcal{P}$ there are $\alpha, \beta \in \mathbb{F}_q^*$ with $\Theta_1 \neq 0$ such that $V_{\alpha, \beta}^{Q+1} \sim_{\text{CCZ}} P$ (in fact can be strengthened to $V_{\alpha, \beta}^{Q+1} \sim_L P$).

Again, we perform Magma experiments to identify these equivalences. For $(k, \ell) \in \{(3, 1), (5, 1), (5, 2)\}$, let S denote the set of representatives of CCZ-equivalence classes of closed butterflies $V_{\alpha, \beta}^{Q+1}$ for $\alpha, \beta \in \mathbb{F}_q^*$ and $\Theta_1 \neq 0$. Then, we identify each $f \in S$ (and $P \in \mathcal{P}$ if necessary) with a polynomial with coefficients in $\mathbb{F}_{q^2}[X]$ through the natural bijection $T : \mathbb{F}_q \times \mathbb{F}_q \rightarrow \mathbb{F}_{q^2}, (x, y) \mapsto x + \tau y$ for a fixed primitive element τ of $\mathbb{F}_{q^2}$. We construct the corresponding linear codes C_f and C_P of length $q^2 - 1$ over $\mathbb{F}_{q^2}$, where C_f has parity check matrix

$$\begin{bmatrix} \cdots & 1 & \cdots \\ \cdots & x & \cdots \\ \cdots & f(x) & \cdots \end{bmatrix}$$

and similarly for C_P . Due to the limitation of the Magma program, we instead consider the binary linear code $\tilde{C}_f$ (and $\tilde{C}_P$) whose parity-check matrix is given by the above matrix with each entry of the last two rows replaced by a column vector being the binary coordinates with respect to a fixed $\mathbb{F}_2$ -basis of $\mathbb{F}_{q^2}$. It is known that f and P are CCZ-equivalent if and only if C_f and C_P (or $\tilde{C}_f$ and $\tilde{C}_P$) are code equivalent [2]. The experimental results are shown in Table 1, where θ is a root of $x^3 + x + 1 \in \mathbb{F}_2[x]$

for $k = 3$ and a root of $x^5 + x^2 + 1 \in \mathbb{F}_2[x]$ for $k = 5$ (in the latter case, $\theta = g^{33}$ where g is the primitive element of $\mathbb{F}_{2^{10}}^*$ as defined in [23, Section 4.4]).

$(k, \ell) = (3, 1)$		$(k, \ell) = (5, 1)$		$(k, \ell) = (5, 2)$	
Function	Representative (α, β)	Function	Representative (α, β)	Function	Representative (α, β)
$\mathcal{G}$	$(1, 1)$	$\mathcal{G}$	$(1, 1)$	$\mathcal{G}$	$(1, 1)$
F_1	$(1, \theta^3)$	F_1	$(\theta^5, 1)$	F_1	$(\theta^{15}, 1)$
F_3	$(\theta, 1)$	F_3	$(\theta, 1)$	F_3	$(\theta^5, 1)$
f_1	$(\theta^3, 1)$	F_5	$(1, \theta)$	F_5	$(1, \theta^3)$
f_3	$(1, \theta)$	F_7	(θ, θ^{22})	F_7	(θ^3, θ^5)
		F_{11}	(θ, θ)	F_{11}	(θ, θ^4)
		F_{15}	$(\theta^{15}, 1)$	F_{15}	$(1, \theta^{15})$
		f_1	$(1, \theta^5)$	f_1	$(\theta^{11}, 1)$
		f_3	(θ, θ^{21})	f_3	$(\theta, 1)$
		f_5	$(\theta^3, 1)$	f_5	$(\theta^3, 1)$
		f_7	$(\theta^7, 1)$	f_7	(θ, θ^6)
		f_{11}	$(\theta^{11}, 1)$	f_{11}	(θ, θ^3)
		f_{15}	(θ, θ^5)	f_{15}	$(1, \theta^{11})$

Table 1 CCZ-representatives of functions in $\mathcal{P}$ in the form $V_{\alpha, \beta}^{Q+1}(x, y)$, $(k, \ell) \in \{(3, 1), (5, 1), (5, 2)\}$

It was proved in [7] that $V_{\alpha, \beta}^{Q+1}$ is APN if and only if $k = 3$, $\text{Tr}_1^3(\alpha) = 0$ and $\beta \in \{\alpha^{Q-1} + \alpha^{Q+1}, \alpha^{-Q+1} + \alpha^{Q+1}\}$. Taking $\ell = 1$ and $\alpha = \theta$, by Table 1, we see that this function is CCZ-equivalent to F_3 . It is also known to be linear (and hence CCZ-) equivalent to the (univariate) Kim function $\kappa : \mathbb{F}_{2^6} \rightarrow \mathbb{F}_{2^6}, X \mapsto X^3 + X^{10} + uX^{24}$, where u is a root of $X^6 + X^4 + X^3 + X + 1 \in \mathbb{F}_2[X]$.

The answers to the following problems are still open:

1. Can the condition $(Q + 1)^k \not\equiv 1 \pmod{R}$ be dropped or weakened in Statement 2(b) of Theorem 2 in general?
2. Is it true that each $P \in \mathcal{P}$ is CCZ-equivalent to some closed butterfly $V_{\alpha, \beta}^{Q+1} \in \mathcal{S}$ for odd $k \geq 7$?
3. Are there any general explicit coefficient conditions that further determine the precise $i \in \mathcal{C}$ for the two cases in Statement 1(b) of Theorem 2?

1.4 Organization of the paper

The paper is now organized as follows. In Section 2 we introduce the universal notations and several basic and necessary concepts used in this paper. In Section 3 we state and prove a general coefficient condition for generalized closed butterflies constructed under biprojective functions to be a permutation. In Section 4 we prove the main theorem (that is, Theorem 2). This part is further divided into three subsections. In Subsection 4.1 we first prove a finer equivalence result with relation to the coefficients of the butterfly structure (see Theorem 13). We adopt a similar method to [10]. In Subsection 4.2 we further classify the EA-equivalence based on the techniques in [13]. In Subsection 4.3 we study the CCZ-equivalence to complete the proof of Theorem 2.

2 Preliminaries

2.1 Notation

The following notations are used throughout this paper:

- $\mathbb{F}_q$ denotes the finite field with q elements and $p := \text{char}(\mathbb{F}_q)$;
- For any field K , $\overline{K}$ denotes the algebraic closure of K ;
- For any field K , $\mathbb{P}^1(K) := K \cup \{\infty\}$;
- For any positive integer n denote by μ_n the set of n -th roots of unity in $\overline{\mathbb{F}_q}$;
- For positive integers m, n such that $m \mid n$, and $\alpha \in \overline{\mathbb{F}_q}$, $\text{Tr}_m^n(\alpha) := \sum_{i=0}^{n/m-1} \alpha^{p^{im}}$.
In particular, if $\alpha \in \mathbb{F}_{p^n}$, then $\text{Tr}_m^n(\alpha)$ is simply the trace of α from $\mathbb{F}_{p^n}$ to $\mathbb{F}_{p^m}$;
- For $\alpha \in \mathbb{F}_{q^2}$ denote $\bar{\alpha} := \alpha^q$;
- For a polynomial $f(x) = \sum_{i=0}^n a_i x^i \in \overline{\mathbb{F}_q}[x]$ and a positive integer m , denote $f^{(m)}(x) := \sum_{i=0}^n a_i^m x^i$.
- For any finite set A , denote $\sharp A$ to be the number of elements of A ;
- $\mathcal{C}$ denotes the set of nonzero 2-cyclotomic coset representatives modulo $(q-1)$.

2.2 Some elementary number theory results

The following elementary number theory result is used throughout this paper.

Lemma 3 ([18, Lemma 4.3]) *Let p be a prime, k, ℓ be positive integers and $e = \gcd(k, \ell)$.*

Then

$$\gcd(p^k + 1, p^\ell + 1) = \begin{cases} 1 & (p = 2 \text{ and } k\ell/e^2 \text{ even}) \\ 2 & (p \geq 3 \text{ and } k\ell/e^2 \text{ even}) \\ p^e + 1 & (k\ell/e^2 \text{ odd}); \end{cases}$$

$$\gcd(p^k + 1, p^\ell - 1) = \begin{cases} 1 & (p = 2 \text{ and } \ell/e \text{ odd}) \\ 2 & (p \geq 3 \text{ and } \ell/e \text{ odd}) \\ p^e + 1 & (\ell/e \text{ even}). \end{cases}$$

2.3 Self-Reciprocal (SR) polynomials

Given a polynomial $f(x) = \sum_{i=0}^d a_i x^i \in \mathbb{F}_q[x]$ where $a_d \neq 0$, its *reciprocal polynomial* $\hat{f}(x)$ is defined by $\hat{f}(x) := \sum_{i=0}^d a_{d-i} x^i = x^d f(x^{-1})$.

We say that a nonzero $f(x) \in \mathbb{F}_q[x]$ is *self-reciprocal* (SR for short) if $\hat{f}(x) = \alpha f(x)$ for some $\alpha \in \mathbb{F}_q^*$.

It is clear that reciprocal polynomials and SR polynomials satisfy the following:

Lemma 4 *All of the following hold:*

- if $f(x) \in \mathbb{F}_q[x]$ is nonzero and $\alpha \in \overline{\mathbb{F}}_q^*$, then the multiplicity of α as a root of $f(x)$ equals the multiplicity of α^{-1} as a root of $\hat{f}(x)$;
- $f(x) \in \mathbb{F}_q[x]$ is SR if and only if the multiset of roots of $f(x)$ is preserved by the inverse function $\alpha \mapsto \alpha^{-1}$;
- if $\alpha \in \mathbb{F}_q^*$ and $\beta \in \mathbb{F}_q$ then $\alpha x^2 + \beta x + \alpha$ is SR.

Lemma 5 Let $p = 2$ and $\alpha, \beta \in \mathbb{F}_q^*$. Then the polynomial $\alpha x^2 + \beta x + \alpha$ has two distinct roots in $\mathbb{F}_{q^2}^*$, which are inverses of each other. Moreover, the roots are in $\mathbb{F}_q^*$ if and only if $\text{Tr}_1^k(\frac{\alpha}{\beta}) = 0$.

2.4 Projective and biprojective functions

Let $q = p^k$ and $Q = p^\ell$ for some prime number p and positive integers k, ℓ . A Q -biprojective polynomial is a bivariate polynomial of the form

$$R(x, y) = a_0 x^{Q+1} + a_1 x^Q y + a_2 x y^Q + a_3 y^{Q+1} \in \mathbb{F}_q[x, y]. \quad (2)$$

The right side of (2) can be shorthand denoted by $(a_0, a_1, a_2, a_3)_Q$.

Taking $y = 1$, then the (univariate) polynomial

$$R(x) := R(x, 1) = a_0 x^{Q+1} + a_1 x^Q + a_2 x + a_3 \in \mathbb{F}_q[x]$$

is called a Q -projective polynomial.

A (Q, Q) -biprojective function is a duple of Q -biprojective polynomials, that is,

$$F(x, y) = (R(x, y), P(x, y)), \quad (3)$$

where

$$P(x, y) = (b_0, b_1, b_2, b_3)_Q \in \mathbb{F}_q[x, y].$$

Remark 1 More generally, we can define a (Q, Q') -biprojective function

$$F(x, y) = ((a_0, a_1, a_2, a_3)_Q, (b_0, b_1, b_2, b_3)_{Q'}) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y],$$

where $q = p^k$, $Q = p^\ell$ and $Q' = p^{\ell'}$ for some positive integers k, ℓ, ℓ' .

Assume $P(x, y)$ is not the zero polynomial. The corresponding (univariate) rational function

$$h(x) = \frac{R(x)}{P(x)} := \frac{R(x, 1)}{P(x, 1)} = \frac{a_0x^{Q+1} + a_1x^Q + a_2x + a_3}{b_0x^{Q+1} + b_1x^Q + b_2x + b_3} \in \mathbb{F}_q(x) \quad (4)$$

is a Q -projective rational function.

Let $c(x) := \gcd(R(x), P(x))$, so that

$$R(x) = c(x)R_0(x), \quad P(x) = c(x)P_0(x)$$

with $\gcd(R_0(x), P_0(x)) = 1$. Note that $c(x)$, $R_0(x)$ and $P_0(x)$ are not necessarily Q -projective polynomials in general.

Then we identify $h(x)$ with $\frac{R_0(x)}{P_0(x)}$, so that h is well-defined as a function from $\mathbb{P}^1(\overline{\mathbb{F}}_q)$ to itself.

The *degree* of the rational function h is defined as $\deg h = \max\{\deg R_0, \deg P_0\}$.

2.5 Relation between biprojective functions and DO polynomials

Let F be the (Q, Q) -biprojective function (3). Define a simple linear bijection

$$\begin{aligned} T : \mathbb{F}_q \times \mathbb{F}_q &\rightarrow \mathbb{F}_{q^2} \\ (x, y) &\mapsto x + \tau y \end{aligned} \quad (5)$$

where τ is a fixed primitive element of $\mathbb{F}_{q^2}^*$. It can be easily verified that the map $f : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_{q^2}$ defined by $f(X) := T \circ F \circ T^{-1}(X)$ can be written in the form

$$f(X) = c_0\bar{X}^{Q+1} + c_1\bar{X}^Q X + c_2\bar{X}X^Q + c_3X^{Q+1}, \quad (6)$$

where $c_0, c_1, c_2, c_3 \in \mathbb{F}_{q^2}$ and $\bar{X} := X^q$.

Note that (6) belongs to the class of *Dembowski-Ostrom (DO) polynomials*, which are polynomials in the form

$$\sum_{0 \leq i \leq j \leq k-1} a_{ij} x^{p^i + p^j}. \quad (7)$$

Moreover, (6) can be rewritten as

$$f(X) = X^{Q+1} A(X^{q-1}), \quad A(X) := c_0 X^{Q+1} + c_1 X^Q + c_2 X + c_3 \in \mathbb{F}_{q^2}[X]. \quad (8)$$

Note that $A(X)$ is a Q -projective polynomial over $\mathbb{F}_{q^2}$.

Now define $B(X) := X^{Q+1} A^{(q)}(1/X) = \bar{c}_3 X^{Q+1} + \bar{c}_2 X^Q + \bar{c}_1 X + \bar{c}_0$. It was shown in [10, Lemma 2.10] that if $A(X)$ has no roots in μ_{q+1} , then the rational function

$$g(X) := \frac{B(X)}{A(X)} \quad (9)$$

induces the same function as $g_0(X) := X^{Q+1} A(X)^{q-1}$ on μ_{q+1} .

The AGW criterion [1] says that f is a permutation polynomial over $\mathbb{F}_{q^2}$ if and only if $\gcd(Q+1, q-1) = 1$ and $g_0(X)$ permutes μ_{q+1} . Ding and Zieve [10] used this criterion and tools for studying the geometric properties of the rational function g (9) to derive necessary and sufficient conditions on the coefficients of f (6) for it to be a permutation over $\mathbb{F}_{q^2}$. In fact they proved for a more general class of polynomials, in the form $X^r A(X^{q-1})$ where $r \equiv Q+1 \pmod{q+1}$. Restricted to the setting of (6), the results in [10, Theorem 1.1] can be restated as follows:

Theorem 6 *Let $q = p^k$ and $Q = p^\ell$ for some positive integers k, ℓ . Let the polynomial $f : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_{q^2}$ be defined as in (6). Define the following coefficient parameters:*

$$\begin{aligned} \theta_1 &= c_0 \bar{c}_0 + c_1 \bar{c}_1 + c_2 \bar{c}_2 + c_3 \bar{c}_3 \\ \theta_2 &= \bar{c}_0 c_1 + \bar{c}_2 c_3 \end{aligned}$$

$$\theta_3 = \bar{c}_0 c_2 + \bar{c}_1 c_3$$

$$\theta_4 = c_1 \bar{c}_1 + c_2 \bar{c}_2.$$

Then f permutes $\mathbb{F}_{q^2}$ if and only if all of the following hold:

- $\gcd(Q+1, q-1) = 1$;
- $\theta_1 \neq 0$;
- $\theta_2^Q = \theta_1^{Q-1} \theta_3$; and
- $\text{Tr}_{2^m}^k(\frac{\theta_4}{\theta_1}) = \frac{\text{lcm}(k, \ell)}{2^m}$, where m is the largest positive integer such that $2^m \mid \gcd(k, \ell)$.

Here the first condition in Theorem 6 is equivalent to $p = 2$ and k/e being odd where $e := \gcd(k, \ell)$ by Lemma 3. This in particular implies that $\text{lcm}(k, \ell)$ has the same 2-adic valuation as ℓ . Moreover, from the proof in [10] it can be easily seen that the result is unaffected if 2^m in the last condition is replaced by e . Hence the last condition can be rephrased as $\text{Tr}_e^k(\frac{\theta_4}{\theta_1}) = \frac{\ell}{e}$.

On the other hand, from [8, Lemma 15], we see that the rational functions $h \in \mathbb{F}_q(x)$ defined in (4) and $g \in \mathbb{F}_{q^2}(X)$ defined in (9) are related by $g = \rho^{-1} \circ h \circ \sigma$ for some degree-one $\rho, \sigma \in \mathbb{F}_{q^2}(X)$ which map μ_{q+1} onto $\mathbb{P}^1(\mathbb{F}_q)$. Hence $\deg h = \deg g$, and the geometric structures of h and g (as maps over $\mathbb{P}^1(\overline{\mathbb{F}_q})$) are the same. Moreover, g permutes μ_{q+1} if and only if h permutes $\mathbb{P}^1(\mathbb{F}_q)$. Combining all, then we can say that F permutes $\mathbb{F}_q \times \mathbb{F}_q$ if and only if $p = 2, k/e$ is odd and h permutes $\mathbb{P}^1(\mathbb{F}_q)$, which is the method used in [15]. In particular, they proved that all (Q, Q) -biprojective permutation functions F over $\mathbb{F}_q \times \mathbb{F}_q$ (2) are linear equivalent to either a Gold function over $\mathbb{F}_{q^2}$ or a “doubly-Gold” function over $\mathbb{F}_q \times \mathbb{F}_q$. A little thought on the results written in [10, Theorem 1.2] restricted to the case $r = Q + 1$ also gives this linear equivalence result (see [5] for more details).

2.6 Ramification and branch points of rational functions

For a non-constant rational function $G(X) \in \overline{\mathbb{F}}_q(X)$, write $G(X) = N(X)/D(X)$ where $N(X), D(X) \in \overline{\mathbb{F}}_q[X]$ with $\gcd(N(X), D(X)) = 1$. For any $\alpha \in \overline{\mathbb{F}}_q$, define $H_\alpha(X) \in \overline{\mathbb{F}}_q[X]$ as

$$H_\alpha(X) := \begin{cases} N(X) - G(\alpha)D(X) & (G(\alpha) \in \overline{\mathbb{F}}_q), \\ D(X) & (G(\alpha) = \infty). \end{cases}$$

It is clear that $H_\alpha(\alpha) = 0$ for all $\alpha \in \overline{\mathbb{F}}_q$. The *ramification index* $e_G(\alpha)$ of α is then its multiplicity as a root of $H_\alpha(X)$. The ramification index of ∞ is defined as $e_G(\infty) := e_{G_1}(0)$, where $G_1(X) := G(\frac{1}{X})$. Given any $\beta \in \mathbb{P}^1(\overline{\mathbb{F}}_q)$, the *ramification multiset* $E_G(\beta)$ of $G(X)$ over β is the multiset of ramification indices $e_G(\alpha)$ for any $\alpha \in G^{-1}(\beta)$, that is, $E_G(\beta) := [e_G(\alpha) : \alpha \in G^{-1}(\beta)]$. In particular, the elements in $E_G(\beta)$ are positive integers whose sum is $\deg G$. We call $\alpha \in \mathbb{P}^1(\overline{\mathbb{F}}_q)$ a *ramification point* (or *critical point*) of $G(X)$ if $e_G(\alpha) > 1$, and its corresponding image $G(\alpha)$ a *branch point* (or *critical value*) of G . Hence a point $\beta \in \mathbb{P}^1(\overline{\mathbb{F}}_q)$ is a branch point of G if and only if $E_G(\beta) \neq [1^{\deg G}]$, where $[m^n]$ denotes the multiset consisting of n copies of m , or equivalently, $\#G^{-1}(\beta) < \deg G$.

2.7 Equivalences between functions over finite fields

In this section we recall the several kinds of equivalences commonly studied in the literature for functions from a finite field to itself.

Definition 1 • A mapping L from $\mathbb{F}_q$ to itself is **linear** if $L(x + y) = L(x) + L(y)$ for

all $x, y \in \mathbb{F}_q$;

- A mapping A from $\mathbb{F}_q$ to itself is **affine** if $A(x) = L(x) + c$ for some linear mapping L and constant $c \in \mathbb{F}_q$;

- The **graph** of a function f from $\mathbb{F}_q$ to itself is the set $\Gamma_f := \{(x, f(x)) : x \in \mathbb{F}_q\}$.

In particular, a polynomial over $\mathbb{F}_q$ is linear if and only if it can be expressed in the form $L(x) = \sum_{i=0}^{k-1} a_i x^{p^i}$.

Definition 2 Let f, g be functions from $\mathbb{F}_q$ to itself. We say f, g are:

- **linear equivalent** if there are linear permutations L and M of $\mathbb{F}_q$ such that $g = L \circ f \circ M$;
- **affine equivalent** if there are affine permutations A and B of $\mathbb{F}_q$ such that $g = A \circ f \circ B$;
- **extended linear (EL-) equivalent** if there are linear permutations L, M and a linear mapping N over $\mathbb{F}_q$ such that $g = L \circ f \circ M + N$;
- **extended affine (EA-) equivalent** if there are affine permutations A, B and an affine mapping P over $\mathbb{F}_q$ such that $g = A \circ f \circ B + P$;
- **Carlet-Charpin-Zinoviev (CCZ-) equivalent** if there is an affine permutation $\mathcal{A}$ of $\mathbb{F}_q \times \mathbb{F}_q$ such that $\Gamma_g = \mathcal{A}(\Gamma_f)$.

CCZ-equivalence can be restated as follows: f, g are CCZ-equivalent if and only if there are linear mappings M, N, K, L over $\mathbb{F}_q$, a permutation π of $\mathbb{F}_q$ and constant elements $\alpha, \beta \in \mathbb{F}_q$ such that $\begin{pmatrix} M & K \\ N & L \end{pmatrix} \in \text{GL}(\mathbb{F}_q \times \mathbb{F}_q)$ and

$$\begin{pmatrix} \pi(x) \\ g \circ \pi(x) \end{pmatrix} = \begin{pmatrix} M & K \\ N & L \end{pmatrix} \begin{pmatrix} x \\ f(x) \end{pmatrix} + \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Then we can easily see from the definitions that EA-equivalence is simply CCZ-equivalence with $K = 0$, EL-equivalence is EA-equivalence with $\alpha = \beta = 0$, and affine (resp. linear) equivalence is EA- (resp. EL-) equivalence with $N = 0$.

It is well-known that Walsh and differential spectra are invariant under CCZ-equivalence. From the above we see that they are also invariant under all the equivalence relations in Definition 2. In particular APN and planar, that is, having differential uniformity 2 and 1 respectively, are invariant properties under all these equivalences. While these equivalences are in general different, it was proved that CCZ-equivalence coincides with EA-equivalence for quadratic APN functions [25, Theorem 1] and general planar functions [19]. For quadratic functions fixing 0, EA-equivalence is the same as EL-equivalence [20, Proposition 2.2]. For planar DO polynomials even all equivalences are the same [3, 4]. These coincidences are the basis of the classifications in [13, 16] and [8] respectively.

One parameter that is invariant under EA-equivalence but not CCZ-equivalence in general is the algebraic degree, which shows that these two equivalences are indeed different. On the other hand, the permutation property is preserved only up to affine equivalence. For example, linear permutations are EL-equivalent to the constant zero function (which is clearly not a permutation).

Now define $\mathcal{L}$ to be the collection of maps from $\mathbb{F}_q \times \mathbb{F}_q$ to itself given by

$$\mathcal{L} := \{(x, y) \mapsto (ax + by, cx + dy) : a, b, c, d \in \mathbb{F}_q, ad \neq bc\}. \quad (10)$$

It can be easily seen that all the maps in $\mathcal{L}$ are linear permutations on $\mathbb{F}_q \times \mathbb{F}_q$.

For (Q, Q) -biprojective functions F, G , we denote $F \sim_{\mathcal{L}} G$ if $F = L \circ G \circ M$ for some $L, M \in \mathcal{L}$. It can be easily checked that $\sim_{\mathcal{L}}$ is an equivalence relation (we call it $\mathcal{L}$ -equivalence), which is finer than linear equivalence in general.

Due to the identification of (Q, Q) -biprojective functions with DO polynomials of the form (6), we may define $\mathcal{L}$ -equivalence between a DO polynomial of the form (6) and a (Q, Q) -biprojective function, or between two DO polynomials of the form (6)

naturally. That is, if f_1, f_2 are DO polynomials of the form (6), then $f_1 \sim_{\mathcal{L}} F$ and $f_1 \sim_{\mathcal{L}} f_2$ mean $T^{-1} \circ f_1 \circ T \sim_{\mathcal{L}} F$ and $T^{-1} \circ f_1 \circ T \sim_{\mathcal{L}} T^{-1} \circ f_2 \circ T$ respectively.

In other words, $f_1 \sim_{\mathcal{L}} f_2$ if and only if $f_1 = L_0 \circ f_2 \circ M_0$ for some $L_0, M_0 \in T\mathcal{L}T^{-1}$, where

$$T\mathcal{L}T^{-1} := \{T \circ L \circ T^{-1} : L \in \mathcal{L}\}.$$

With some computation, we may further rewrite this set as

$$T\mathcal{L}T^{-1} = \{M(X) = \alpha\bar{X} + \beta X : \alpha, \beta \in \mathbb{F}_{q^2}, \alpha^{q+1} \neq \beta^{q+1}\}.$$

We now summarize and rewrite [8, Lemma 15 and Theorem 16] (with the proofs of Lemmas 12–13) in terms of $\mathcal{L}$ -equivalence as follows:

Lemma 7 ([8, Lemma 15]) *Let $q = p^k$ and $Q = p^\ell$ for some positive integers k, ℓ . For $\underline{c} = (c_0, c_1, c_2, c_3) \in \mathbb{F}_{q^2}^4 \setminus \{(0, 0, 0, 0)\}$, define $A_{\underline{c}}(X) = c_0X^{Q+1} + c_1X^Q + c_2X + c_3 \in \mathbb{F}_{q^2}[X]$. Let $B_{\underline{c}}(X) = X^{Q+1}A_{\underline{c}}^{(q)}(1/X)$, $f_{\underline{c}}(X) = X^{Q+1}A_{\underline{c}}(X^{q-1})$ and $g_{\underline{c}}(X) = B_{\underline{c}}(X)/A_{\underline{c}}(X)$. Assume $\deg g_{\underline{c}} = Q + 1$.*

1. $f_{\underline{c}} \sim_{\mathcal{L}} (P_1(x, y), P_2(x, y))$ where P_1, P_2 are given Q -biprojective polynomials if and only if $g_{\underline{c}}(X) = \rho^{-1} \circ \frac{P_1(X, 1)}{P_2(X, 1)} \circ \sigma$ for some degree-one $\rho, \sigma \in \mathbb{F}_{q^2}(X)$ both of which map μ_{q+1} to $\mathbb{P}^1(\mathbb{F}_q)$;
2. $f_{\underline{c}} \sim_{\mathcal{L}} f_{\underline{c}_0}$ for $\underline{c}, \underline{c}_0 \in \mathbb{F}_{q^2}^4$ if and only if $g_{\underline{c}} = \rho^{-1} \circ g_{\underline{c}_0} \circ \sigma$ for some degree-one $\rho, \sigma \in \mathbb{F}_{q^2}(X)$ both of which permute μ_{q+1} .

Proposition 8 ([8, Theorem 16, Proof of Lemmas 12–13]) *Let $q = p^k$ and $Q = p^\ell$ for some positive integers k, ℓ . Let $A(X) = c_0X^{Q+1} + c_1X^Q + c_2X + c_3 \in \mathbb{F}_{q^2}[X]$ be a nonzero polynomial, $B(X) = X^{Q+1}A^{(q)}(1/X)$, $f(X) = X^{Q+1}A(X^{q-1})$ and $g(X) = B(X)/A(X)$. Assume $g(X)$ is nonconstant and $A(X)$ has no roots in μ_{q+1} . Then f is $\mathcal{L}$ -equivalent to one of the following polynomials over $\mathbb{F}_{q^2}$ or (Q, Q) -biprojective functions over $\mathbb{F}_q \times \mathbb{F}_q$:*

1. $(x^{Q+1}, y^{Q+1}) : \mathbb{F}_q \times \mathbb{F}_q \rightarrow \mathbb{F}_q \times \mathbb{F}_q$;

2. X^{Q+1} ;
3. X^{Q+q} ;
4. $(x^{Q+1}, xy^Q + y^{Q+1}) : \mathbb{F}_q \times \mathbb{F}_q \rightarrow \mathbb{F}_q \times \mathbb{F}_q$;
5. $(x^Q y, x^{Q+1} + \varepsilon y^{Q+1}) : \mathbb{F}_q \times \mathbb{F}_q \rightarrow \mathbb{F}_q \times \mathbb{F}_q$ for some $\varepsilon \in \mathbb{F}_q^*$;
6. $(x^{Q+1} - x^Q y, xy^Q + \varepsilon y^{Q+1}) : \mathbb{F}_q \times \mathbb{F}_q \rightarrow \mathbb{F}_q \times \mathbb{F}_q$ for some $\varepsilon \in \mathbb{F}_q^* \setminus \{-1\}$;
7. $X^{Q+q} + \varepsilon X^{Q+1}$ for some $\varepsilon \in \mathbb{F}_{q^2}^* \setminus \mu_{q+1}$.

The corresponding nature of branch points and ramification of g are as follows:

1. $\deg g = Q + 1$, g has 2 ramification points in μ_{q+1} , both of which are unique preimages of a corresponding branch point in μ_{q+1} ;
2. $\deg g = Q + 1$, g has 2 ramification points outside μ_{q+1} , both of which are unique preimages of a corresponding branch point outside μ_{q+1} ;
3. $\deg g = Q - 1$, g has 2 ramification points outside μ_{q+1} , both of which are unique preimages of a corresponding branch point outside μ_{q+1} ;
4. g has 2 branch points in μ_{q+1} , of ramification multisets $[Q + 1]$ and $[1, Q]$ respectively;
5. g has a unique branch point (in μ_{q+1}), whose ramification multiset is $[1, Q]$;
6. g has 2 branch points in μ_{q+1} , both of ramification multiset $[1, Q]$;
7. g has 2 branch points outside μ_{q+1} , both of ramification multiset $[1, Q]$.

3 Permutation criterion for generalized closed butterflies

Let $R(x, y)$ be a Q -biprojective polynomial over $\mathbb{F}_q$. We recall that the closed butterfly with respect to R is

$$\mathbb{V}_R(x, y) = (R(x, y), R(y, x)) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y].$$

Note that here we do not need to make the assumption that $R_y: x \mapsto R(x, y)$ permutes $\mathbb{F}_q$ for any fixed $y \in \mathbb{F}_q$ to make V_R well-defined.

The following result states necessary and sufficient coefficient conditions such that V_R defines a permutation on $\mathbb{F}_q \times \mathbb{F}_q$. Note the similarity of the conditions with those in Theorem 6.

Theorem 9 *Let $q = p^k$ and $Q = p^\ell$ for some positive integers k, ℓ , and denote $e = \gcd(k, \ell)$. Let $R(x, y) = a_0x^{Q+1} + a_1x^Qy + a_2xy^Q + a_3y^{Q+1} \in \mathbb{F}_q[x, y]$ be a Q -biprojective polynomial. Define the coefficient parameters*

$$\Theta_1 := a_0^2 + a_1^2 + a_2^2 + a_3^2$$

$$\Theta_2 := a_0a_1 + a_2a_3$$

$$\Theta_3 := a_0a_2 + a_1a_3$$

$$\Theta_4 := a_1^2 + a_2^2$$

Then V_R permutes $\mathbb{F}_q \times \mathbb{F}_q$ if and only if all of the following hold:

- $p = 2$ and k/e is odd;
- $\Theta_1 \neq 0$;
- $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$; and
- $\text{Tr}_e^k(\frac{\Theta_4}{\Theta_1}) = 0$.

We remark that by the linear bijection T (5), the functions in Theorem 9 can be transformed into polynomials over $\mathbb{F}_{q^2}$ of the form (6). In particular, one may use Theorem 6 directly to prove Theorem 9, which involves technical computations to change the conditions from one to another. Here we provide an alternative proof, which is essentially applying Ding and Zieve's techniques in [10] to (Q, Q) -biprojective functions without computing the explicit coefficients of the corresponding polynomials

(6) over $\mathbb{F}_{q^2}$. Before doing so, we first state three results, which are direct consequences of those from [10] restricted to the case that all involved coefficients are in $\mathbb{F}_q$.

Lemma 10 ([10, Theorem 3.1]) *Let $q = 2^k$ and $Q = 2^\ell$ for some positive integers k, ℓ . Let $R(x) = a_0x^{Q+1} + a_1x^Q + a_2x + a_3 \in \mathbb{F}_q[x]$ with $R(1) \neq 0$. Let $P(x) = a_3x^{Q+1} + a_2x^Q + a_1x + a_0$ and $h(x) := \frac{R(x)}{P(x)}$. Define the coefficient parameters*

$$\Theta_1 := a_0^2 + a_1^2 + a_2^2 + a_3^2$$

$$\Theta_2 := a_0a_1 + a_2a_3$$

$$\Theta_3 := a_0a_2 + a_1a_3.$$

Then the following holds:

1. $\deg h \in \{Q+1, Q-1\}$;
2. $\deg h = Q-1$ if and only if $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$, $\{a_1, a_2, a_3\} \neq \{0\}$ and $P(x)$ (resp. $R(x)$) is divisible by $u(x) := \Theta_2x^2 + \Theta_1x + \Theta_2$;
3. Assume $\deg h > 1$. Then h has two distinct ramification points in $\mathbb{P}^1(\overline{\mathbb{F}}_q)$ (in fact in $\mathbb{P}^1(\mathbb{F}_{q^2})$), which are the roots of $v(x) := \Theta_3^{1/Q}x^2 + \Theta_1^{1/Q}x + \Theta_3^{1/Q}$ if $\Theta_3 \neq 0$, or $0, \infty$ if $\Theta_3 = 0$. The corresponding branch points are the roots of $w(x) := (a_0a_3 + a_1a_2)x^2 + \Theta_1x + (a_0a_3 + a_1a_2)$ if $a_0a_3 \neq a_1a_2$, or $0, \infty$ otherwise;
4. Assume $\deg h > 1$. Then the ramification points of h are unique preimages of the corresponding branch points if and only if $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$, otherwise the ramification multisets for both branch points are $[1, Q]$.

Proof Most arguments follow from the proof of [10, Theorem 3.1]. Some extra remarks are as follows: since $\Theta_1 = R(1)^2 \neq 0$, 1 is not a root of u, v or w . Then the roots of u, v can be represented in the form $\{\alpha, \alpha^{-1}\}$ and $\{\beta, \beta^{-1}\}$ for some $\alpha, \beta \in \overline{\mathbb{F}}_q \setminus \{1\}$ by Lemma 4. This implies that the sets of roots of u and v are either disjoint or equal. These explain why we can rule out the possibility that h has only one branch point or that the two branch points of h have different ramification multisets. $\square$

Lemma 11 ([10, Lemma 6.2]) Let $q = 2^k$ and $Q = 2^\ell$ for some positive integers k, ℓ . Let $R(x) = a_0x^{Q+1} + a_1x^Q + a_2x + a_3 \in \mathbb{F}_q[x]$ with $R(1) \neq 0$. Define the coefficient parameters

$$\Theta_1 := a_0^2 + a_1^2 + a_2^2 + a_3^2$$

$$\Theta_2 := a_0a_1 + a_2a_3$$

$$\Theta_3 := a_0a_2 + a_1a_3$$

$$\Theta_4 := a_1^2 + a_2^2$$

$$\varphi := \frac{\Theta_4}{\Theta_1} + \text{Tr}_1^\ell \left(\frac{\Theta_2}{\Theta_1} \right).$$

Assume $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$. Then $\varphi \in \mathbb{F}_2$. If we further assume $\{a_1, a_2, a_3\} \neq \{0\}$, then $R(x)$ is divisible by $u(x) := \Theta_2x^2 + \Theta_1x + \Theta_2$ if and only if $\varphi = 1$.

Lemma 12 ([10, Lemma 6.5]) Let $q = 2^k$ and $Q = 2^\ell$ for some positive integers k, ℓ , and denote $e := \gcd(k, \ell)$. Let $a_0, a_1, a_2, a_3 \in \mathbb{F}_q$ such that $a_0 + a_1 + a_2 + a_3 \neq 0$. Define

$$\Theta_1 := a_0^2 + a_1^2 + a_2^2 + a_3^2$$

$$\Theta_2 := a_0a_1 + a_2a_3$$

$$\Theta_3 := a_0a_2 + a_1a_3$$

$$\Theta_4 := a_1^2 + a_2^2$$

$$\varphi := \frac{\Theta_4}{\Theta_1} + \text{Tr}_1^\ell \left(\frac{\Theta_2}{\Theta_1} \right).$$

Assume $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$. Then $\text{Tr}_e^k(\frac{\Theta_4}{\Theta_1}) = \frac{k}{e}\varphi + \frac{\ell}{e}\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1})$.

Proof of Theorem 9 We consider the associated polynomial and rational function f and g over $\mathbb{F}_{q^2}$ defined in (6) and (9) respectively.

From [10, Section 5], we know that (6) permutes $\mathbb{F}_{q^2}$ if and only if both of the following hold:

- (I) $\gcd(Q+1, q-1) = 1$; and
- (II) $g(X) = \rho^{-1} \circ X^n \circ \sigma$, where ρ, σ are degree-one rational functions mapping μ_{q+1} to Λ , and $\gcd(n, r) = 1$ with one of the following holds:

- (a) $n = Q + 1, r = q - 1$ and $\Lambda = \mathbb{P}^1(\mathbb{F}_q)$;
- (b) $n = Q + 1, r = q + 1$ and $\Lambda = \mu_{q+1}$; or
- (c) $n = Q - 1, r = q + 1$ and $\Lambda = \mu_{q+1}$.

By Lemma 3, (I) holds if and only if k/e is odd.

As for (II), we first note that from Statement 1 of Lemma 7, $g(X) \in \mathbb{F}_{q^2}(X)$ and $h(x) := \frac{R(x,1)}{R(1,x)} \in \mathbb{F}_q(x)$ are related by $g = \tau^{-1} \circ h \circ \nu$ where τ, ν are degree-one rational functions mapping μ_{q+1} to $\mathbb{P}^1(\mathbb{F}_q)$. In particular, $\deg h = \deg g = n$ and the geometric properties of g and h (treated as maps defined over $\mathbb{P}^1(\overline{\mathbb{F}}_q)$) are the same. The form of g in (II) means that g has two distinct ramification points, each of which is the unique preimage of a corresponding branch point. Hence Lemma 10 implies that $\Theta_1 \neq 0$ and $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$. Moreover, the two ramification points of h are the roots of $u(x) = \Theta_2 x^2 + \Theta_1 x + \Theta_2$ from its proof. We look at the three cases separately as follows:

(a) In this case both ramification points of g are in μ_{q+1} . Equivalently, this means both ramification points of h are in $\mathbb{P}^1(\mathbb{F}_q)$. By Lemma 5, $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 0$. Moreover, $\deg h = Q + 1$. Hence by Lemma 10, either $\{a_1, a_2, a_3\} = \{0\}$ or $u(x) \nmid R(x, 1)$. If the former holds, then $\Theta_1 = a_0^2$ and $\Theta_2 = \Theta_4 = 0$, so that $\varphi = 0$. Otherwise $\{a_1, a_2, a_3\} \neq \{0\}$ and $u(x) \nmid R(x, 1)$, which also implies $\varphi = 0$ by Lemma 11. In either case, we then get $\text{Tr}_e^k(\frac{\Theta_4}{\Theta_1}) = 0$ by Lemma 12. Note that the gcd condition is just the same as (I) in this case.

(b) In this case both ramification points of g are not in μ_{q+1} . Equivalently, this means both ramification points of h are not in $\mathbb{P}^1(\mathbb{F}_q)$. By Lemma 5, $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 1$. Moreover, $\deg h = Q + 1$. Note that $\{a_1, a_2, a_3\} \neq \{0\}$ since $\Theta_2 \neq 0$. Hence Lemma 10 implies that $u(x) \nmid R(x, 1)$, which further yields $\varphi = 0$ by Lemma 11. We then get $\text{Tr}_e^k(\frac{\Theta_4}{\Theta_1}) = \ell/e$ by Lemma 12. Since k/e is odd, by Lemma 3, we easily see that the gcd condition holds if and only if $k\ell/e^2$ is even if and only if ℓ/e is even if and only if $\text{Tr}_e^k(\frac{\Theta_4}{\Theta_1}) = 0$.

(c) Similar to (b), we have $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 1$. However, we now have $\deg h = Q - 1$. Again $\{a_1, a_2, a_3\} \neq \{0\}$. Hence by Lemma 10, $u(x) \mid R(x, 1)$, which further yields $\varphi = 1$ by Lemma 11. We then get $\text{Tr}_e^k(\frac{\Theta_4}{\Theta_1}) = k/e + \ell/e = 1 + \ell/e$ by Lemma 12, since k/e is odd. Thus by Lemma 3, the gcd condition holds if and only if ℓ/e is odd if and only if $\text{Tr}_e^k(\frac{\Theta_4}{\Theta_1}) = 0$. $\square$

4 Proof of Theorem 2

The proof of Theorem 2 is split into three parts. In Section 4.1 we first prove Statement 1, which shows that the closed butterfly structure is linear (actually $\mathcal{L}$ -)equivalent to some simple polynomials or (Q, Q) -biprojective functions under various coefficient conditions. Then in Sections 4.2 and 4.3 we study the EA-equivalence and CCZ-equivalence to prove Parts (a) and (b) of Statement 2 respectively.

4.1 $\mathcal{L}$ -equivalence

For the convenience of readers, we restate Statement 1 of Theorem 2 as follows.

Theorem 13 *Let $q = 2^k$ and $Q = 2^\ell$ for some positive integers k, ℓ , with k odd and $\gcd(k, \ell) = 1$. Assume $\alpha, \beta \in \mathbb{F}_q^*$.*

Define the parameters

$$\Theta_1 := (\alpha + 1)^{2Q+2} + \beta^2$$

$$\Theta_2 := \alpha^{2Q+1} + \alpha + \alpha^Q \beta$$

$$\Theta_3 := \alpha^{Q+2} + \alpha^Q + \alpha\beta.$$

Assume $\Theta_1 \neq 0$.

1. $V_{\alpha, \beta}^{Q+1}$ induces a permutation on $\mathbb{F}_q \times \mathbb{F}_q$ if and only if $\Theta_2^Q = \Theta_1^{Q-1} \Theta_3$. In this case,

- if ℓ is odd, then $V_{\alpha, \beta}^{Q+1} \sim_{\mathcal{L}} X^{Q+q} \in \mathbb{F}_{q^2}[X]$;
- if ℓ is even, then $V_{\alpha, \beta}^{Q+1} \sim_{\mathcal{L}} X^{Q+1} \in \mathbb{F}_{q^2}[X]$.

2. Otherwise,

- if $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 0$, then $V_{\alpha, \beta}^{Q+1} \sim_{\mathcal{L}} (x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1}) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y]$ for some $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2$;
- if $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 1$, then $V_{\alpha, \beta}^{Q+1} \sim_{\mathcal{L}} X^{Q+q} + \varepsilon X^{Q+1} \in \mathbb{F}_{q^2}[X]$ for some $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2$.

By [21, Theorem 2 (1)], if $\Theta_1 \neq 0$, then the condition $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$ in Statement 1 above is equivalent to $\alpha^2 + \alpha\tilde{\beta} + \tilde{\beta}^2 = 1$ where $\tilde{\beta}$ is the unique $(Q+1)$ -th root of β in $\mathbb{F}_q^*$. Their statement additionally assumes ℓ is odd but in fact this condition has not been used in that proof at all. Therefore the first part of Statement 1 of Theorem 13 is essentially [10, Corollary 9.5]. However in the following we will provide an alternative proof.

Before proving Theorem 13, we need the following result, which is a restatement of [23, Lemma 4] using the notations in our paper.

Lemma 14 *Let $q = 2^k$ and $Q = 2^\ell$ for some positive integers k, ℓ with k odd and $\gcd(k, \ell) = 1$. Let $\alpha, \beta \in \mathbb{F}_q^*$. Define the coefficient parameters*

$$\begin{aligned}\Theta_1 &:= (\alpha + 1)^{2Q+2} + \beta^2 \\ \Theta_2 &:= \alpha^{2Q+1} + \alpha + \alpha^Q\beta \\ \Theta_3 &:= \alpha^{Q+2} + \alpha^Q + \alpha\beta.\end{aligned}$$

Assume $\Theta_1 \neq 0$. Then the two polynomials

$$u(x) = \Theta_2x^2 + \Theta_1x + \Theta_2$$

and

$$v(x)^Q = \Theta_3x^{2Q} + \Theta_1x^Q + \Theta_3$$

have no common roots in $\mathbb{F}_q$.

Proof of Theorem 13 We note that

$$R_{\alpha, \beta}^{Q+1}(x, y) = x^{Q+1} + \alpha x^Q y + \alpha^Q xy^Q + (\alpha^{Q+1} + \beta)y^{Q+1}.$$

From this we easily see that the definitions of $\Theta_i, i = 1, 2, 3$ are all consistent with those defined in Section 3.

Case 1: $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$

First, we have

$$\Theta_4 = \alpha^2 + \alpha^{2Q} = \alpha\Theta_2 + \alpha^Q\Theta_3,$$

so that

$$\frac{\Theta_4}{\Theta_1} = \alpha \frac{\Theta_2}{\Theta_1} + \alpha^Q \frac{\Theta_3}{\Theta_1} = \alpha \frac{\Theta_2}{\Theta_1} + \left(\alpha \frac{\Theta_2}{\Theta_1} \right)^Q.$$

This implies $\text{Tr}_1^k(\frac{\Theta_4}{\Theta_1}) = 0$. Moreover, k is odd and $e = \gcd(k, \ell) = 1$, and so all conditions in Theorem 9 are satisfied. Therefore $V_{\alpha, \beta}^{Q+1}$ permutes $\mathbb{F}_q \times \mathbb{F}_q$, or equivalently, the associated polynomial $f \in \mathbb{F}_{q^2}[X]$ (6) permutes $\mathbb{F}_{q^2}$.

On the other hand, $u(x)$ and $v(x)$ are nonzero constant multiples of each other, that is, their set of roots coincides. Since $\Theta_1 \neq 0$, Lemma 14 implies that the roots of $u(x)$ (and also $v(x)$) are not in $\mathbb{F}_q$. Hence the ramification points of h are not in $\mathbb{P}^1(\mathbb{F}_q)$, or equivalently, those of the associated rational function $g \in \mathbb{F}_{q^2}(X)$ (9) are not in μ_{q+1} . By [10], we see that $g = \rho^{-1} \circ X^n \circ \sigma$ where $\rho, \sigma \in \mathbb{F}_{q^2}(X)$ are degree-one rational functions permuting μ_{q+1} , and $n \in \{Q-1, Q+1\}$ where n is such that $\gcd(n, q+1) = 1$. From the proof of Theorem 9, these latter conditions hold simultaneously if and only if $n = Q-1$ with ℓ odd or $n = Q+1$ with ℓ even. Therefore by Proposition 8, $V_{\alpha, \beta}^{Q+1}$ is $\mathcal{L}$ -equivalent to X^{Q+q} or X^{Q+1} in $\mathbb{F}_{q^2}[X]$ according to whether ℓ is odd or even, as desired.

Case 2: $\Theta_2^Q \neq \Theta_1^{Q-1}\Theta_3$

It is immediate from Theorem 9 that $V_{\alpha, \beta}^{Q+1}$ is not a permutation.

On the other hand, by Lemma 10, $\deg h = Q+1$ and both branch points of h are of ramification multiset $[1, Q]$.

If $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 0$, then both ramification points of h are in $\mathbb{P}^1(\mathbb{F}_q)$, so the ramification points of the associated rational function $g \in \mathbb{F}_{q^2}(X)$ (9) are in μ_{q+1} . By Proposition 8, we have $V_{\alpha, \beta}^{Q+1} \sim_{\mathcal{L}} (x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1}) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y]$ for some $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2$.

If $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 1$ instead, then both ramification points of h are outside $\mathbb{P}^1(\mathbb{F}_q)$, so the ramification points of the associated rational function $g(X) \in \mathbb{F}_{q^2}(X)$ are outside μ_{q+1} . By Proposition 8, we have $V_{\alpha, \beta}^{Q+1} \sim_{\mathcal{L}} f_{\varepsilon}(X) := X^{Q+q} + \varepsilon X^{Q+1} \in \mathbb{F}_{q^2}[X]$ for some $\varepsilon \in \mathbb{F}_{q^2}^* \setminus \mu_{q+1}$.

Now we observe that, for any $\gamma \in \mu_{q+1}$, $f_{\varepsilon\gamma} = \gamma^{(1-Q)/2} X \circ f_{\varepsilon} \circ \gamma^{1/2} X$, so $f_{\varepsilon\gamma} \sim_{\mathcal{L}} f_{\varepsilon}$. Since τ is a primitive element of $\mathbb{F}_{q^2}$, we easily see that μ_{q+1} is the cyclic group generated by τ^{q-1} . Moreover, we may write $\varepsilon = \tau^j$ where $1 \leq j \leq q^2-2$ and $q-1 \nmid j$ (since $\varepsilon \notin \mu_{q+1}$). Since

$\gcd(q-1, q+1) = 1$, there is i such that $j + i(q-1) \equiv 0 \pmod{q+1}$. Taking $\gamma = \tau^{i(q-1)} \in \mu_{q+1}$, we have $V_{\alpha, \beta}^{Q+1} \sim_{\mathcal{L}} f_{\varepsilon'}(X) \in \mathbb{F}_{q^2}[X]$ with $\varepsilon' := \varepsilon\gamma = \tau^{j+i(q-1)} \in \mathbb{F}_q^* \setminus \mu_{q+1} = \mathbb{F}_q \setminus \mathbb{F}_2$.

Combining all three cases completes the proof of Theorem 13. $\square$

Remark 2 From the proof in **Case 1**, it is easy to see that for $V_{\alpha, \beta}^{Q+1}$, if $\Theta_1 \neq 0$ and $\Theta_2^Q = \Theta_1^{Q-1}\Theta_3$, then we must have $\text{Tr}_1^k(\frac{\Theta_2}{\Theta_1}) = 1$.

Theorem 15 Let $q = 2^k$ and $Q = 2^\ell$ for some positive integers k, ℓ . Denote by $\mathcal{P}$ the collection

$$\mathcal{P} := \{(x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1}) : \varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2\} \cup \{X^{Q+q} + \varepsilon X^{Q+1} : \varepsilon \in \mathbb{F}_q \setminus \{1\}\} \cup \{X^{Q+1}\}.$$

Then the functions in $\mathcal{P}$ are pairwise $\mathcal{L}$ -inequivalent.

Proof Pick $f_1, f_2 \in \mathcal{P}$. Using the linear bijection T in (5) for identification if necessary, we may write $f_i(X) = a_i \bar{X}^{Q+1} + b_i \bar{X}^Q X + c_i \bar{X} X^Q + d_i X^{Q+1}$ with $a_i, b_i, c_i, d_i \in \mathbb{F}_{q^2}$ for $i = 1, 2$. Assume $f_1 \sim_{\mathcal{L}} f_2$.

Then Statement 2 of Lemma 7 implies that the associated rational functions

$$g_i(X) := \frac{\bar{d}_i X^{Q+1} + \bar{c}_i X^Q + \bar{b}_i X + \bar{a}_i}{a_i X^{Q+1} + b_i X^Q + c_i X + d_i} \in \mathbb{F}_{q^2}(X)$$

over μ_{q+1} satisfy the relation

$$g_2 = \rho \circ g_1 \circ \sigma, \tag{11}$$

where

$$\sigma(X) = \frac{\bar{\beta}X + \bar{\alpha}}{\alpha X + \beta}, \quad \rho(X) = \frac{\bar{\delta}X + \bar{\gamma}}{\gamma X + \delta}$$

are degree-one rational functions over $\mathbb{F}_{q^2}$ which permute μ_{q+1} .

In particular, g_1, g_2 have the same degree as well as geometric pattern. More precisely, their number of ramification points and branch points, as well as their ramification multisets and whether they are in μ_{q+1} or not, are invariant.

By Proposition 8, the following four families have different geometric properties:

(F1) X^{Q+1} ;

- (F2) X^{Q+q} ;
(F3) $(x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1})$ for $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2$;
(F4) $X^{Q+q} + \varepsilon X^{Q+1}$ for $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2$.

Hence if f_1 and f_2 belong to different families above, then they are $\mathcal{L}$ -inequivalent.

It remains to show that any two different members in the same family are also $\mathcal{L}$ -inequivalent. In particular this only needs to be done for (F3) and (F4).

Case 1: f_i are both in (F3), with $\varepsilon = \varepsilon_i$.

The associated rational functions are given by

$$g_i = \pi \circ \frac{x^Q(x+1)}{x+\varepsilon_i} \circ \pi^{-1},$$

where $\pi(x) := \frac{x+\bar{\tau}}{x+\tau}$ is a degree-one rational function over $\mathbb{F}_{q^2}$ which maps $\mathbb{P}^1(\mathbb{F}_q)$ (bijectively) onto μ_{q+1} .

Define $h_i(x) := \frac{x^Q(x+1)}{x+\varepsilon_i} = \pi^{-1} \circ g_i \circ \pi$. If (11) holds, then

$$h_2(x) = \tilde{\rho} \circ h_1 \circ \tilde{\sigma} \tag{12}$$

where

$$\tilde{\rho} := \pi^{-1} \circ \rho \circ \pi, \tilde{\sigma} := \pi^{-1} \circ \sigma \circ \pi.$$

By [10, Lemma 2.11], $\tilde{\rho}, \tilde{\sigma}$ are degree-one rational functions with coefficients in $\mathbb{F}_q$.

Now (12) implies that $\tilde{\rho}$ maps each branch point of h_1 to a branch point of h_2 (of same ramification multiset), and $\tilde{\sigma}$ maps the corresponding preimage(s) of h_2 to the corresponding preimage(s) of h_1 (of same ramification index). That is, either of the following holds:

- (1A) $\tilde{\rho}(0) = 0, \tilde{\rho}(\infty) = \infty, \tilde{\sigma}(0) = 0, \tilde{\sigma}(1) = 1, \tilde{\sigma}(\infty) = \infty, \tilde{\sigma}(\varepsilon_2) = \varepsilon_1$; or
(1B) $\tilde{\rho}(\infty) = 0, \tilde{\rho}(0) = \infty, \tilde{\sigma}(\infty) = 0, \tilde{\sigma}(\varepsilon_2) = 1, \tilde{\sigma}(0) = \infty, \tilde{\sigma}(1) = \varepsilon_1$.

By [9, Lemma 2.7], any three distinct points determine a degree-one rational function. In particular we get $\tilde{\sigma}(x) = x$ for case (1A) and $\tilde{\sigma}(x) = \frac{\varepsilon_2}{x}$ for case (1B) by the first three conditions on $\tilde{\sigma}$. Then the last condition on $\tilde{\sigma}$ easily yields $\varepsilon_1 = \varepsilon_2$ in either case, and so $f_1 = f_2$.

Case 2: f_i are both in (F4), with $\varepsilon = \varepsilon_i$.

The associated rational functions are given by

$$g_i(X) = \frac{X^Q(\varepsilon_i X + 1)}{X + \varepsilon_i}.$$

If (11) holds, then as in the previous case, either of the following holds:

(2A) $\rho(0) = 0, \rho(\infty) = \infty, \sigma(0) = 0, \sigma(\varepsilon_2^{-1}) = \varepsilon_1^{-1}, \sigma(\infty) = \infty, \sigma(\varepsilon_2) = \varepsilon_1$; or

(2B) $\rho(\infty) = 0, \rho(0) = \infty, \sigma(\infty) = 0, \sigma(\varepsilon_2) = \varepsilon_1^{-1}, \sigma(0) = \infty, \sigma(\varepsilon_2^{-1}) = \varepsilon_1$.

By the first three conditions on σ , we get $\sigma(X) = \varepsilon_1^{-1}\varepsilon_2 X$ for case (2A) and $\sigma(X) = \frac{\varepsilon_1^{-1}\varepsilon_2}{X}$ for case (2B). Then the last condition on σ easily yields $\varepsilon_1 = \varepsilon_2$ in either case, and so $f_1 = f_2$. $\square$

Remark 3 Apart from $\mathcal{L}$ -equivalence, it is easy to see that for any polynomial f , f and $f^{(2^t)}$ are also linear equivalent for any non-negative integer t , due to the identity $f^{(2^t)}(X) = X^{2^t} \circ f \circ X^{1/2^t}$, and thereby $\tilde{f} \sim_L f$ if $\tilde{f}$ is a composition of an $\mathcal{L}$ -equivalence of f and $(\cdot)^{(2^t)}$ for non-negative integer t . We will show that the converse is also true (in fact for coarser equivalences, that is, EA- or even CCZ-equivalence) for the functions in $\mathcal{P}$ under certain conditions on k in the following two subsections.

4.2 EA-equivalence

In this section we prove Statement 2(a) of Theorem 2, that is to further classify the functions listed in Theorem 15 into EA-equivalence classes, which gives an upper bound on the number of EA-inequivalent closed butterfly structures $\mathcal{V}_{\alpha, \beta}^{Q+1}$. The main idea is from [13] with slight modifications.

We use the following notation throughout this and next subsection.

- k, ℓ are positive integers, $q = 2^k$ and $Q = 2^\ell$;
- Let r be a primitive prime divisor of $2^k - 1$, that is, a prime dividing $2^k - 1$ but not $2^i - 1$ for $1 \leq i < k$. Zsigmondy's theorem (see [17, Chapter IX, Theorem 8.3]) implies that such a prime exists if $k \neq 1, 6$. Obviously r is odd.

- $\mathcal{R}$ is the unique Sylow r -subgroup of $\mathbb{F}_q^*$, and $R = \# \mathcal{R}$;
- For $a \in \mathbb{F}_q^*$ denote by $m_a \in \text{GL}(\mathbb{F}_q)$ the linear map $x \mapsto ax$.
- For $A, B, C, D \in \text{GL}(\mathbb{F}_q)$ we write diagonal matrices

$$\text{diag}(A, B) = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \in \text{GL}(\mathbb{F}_{q^2})$$

and

$$\text{diag}(A, B, C, D) = \begin{pmatrix} A & 0 & 0 & 0 \\ 0 & B & 0 & 0 \\ 0 & 0 & C & 0 \\ 0 & 0 & 0 & D \end{pmatrix} \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2}).$$

- $\Delta_{\mathcal{R}} := \{\text{diag}(m_a, m_a) : a \in \mathcal{R}\}$;
- $Z := \{\text{diag}(m_a, m_a, m_{a^{Q+1}}, m_{a^{Q+1}}) : a \in \mathbb{F}_q^*\}$;
- $Z_{\mathcal{R}} := \{\text{diag}(m_a, m_a, m_{a^{Q+1}}, m_{a^{Q+1}}) : a \in \mathcal{R}\}$;
- $\text{ELM} := \left\{ \begin{pmatrix} M & 0 \\ N & L \end{pmatrix} \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2}) \right\}$;
- For a (Q, Q) -biprojective function F we denote by:
 - $\text{Aut}_{\text{EL}}(F) := \{\mathcal{A} \in \text{ELM} : \mathcal{A}(\Gamma_F) = \Gamma_F\}$;
 - $\text{Aut}(F) := \{\mathcal{A} \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2}) : \mathcal{A}(\Gamma_F) = \Gamma_F\}$;
 - C_F and C'_F the centralizers of $Z_{\mathcal{R}}$ in $\text{Aut}_{\text{EL}}(F)$ and $\text{Aut}(F)$ respectively (the fact that $Z_{\mathcal{R}}$ is a subgroup of $\text{Aut}_{\text{EL}}(F)$ [and hence also of $\text{Aut}(F)$] is proved in [13, Lemma 3]).

First, since our functions F are always quadratic and fix 0, by [20, Proposition 2.2], EA-equivalence is the same as EL-equivalence.

We first state and prove a modified version of [13, Theorem 3] for (Q, Q) -biprojective functions:

Theorem 16 Assume $k > 2$ and $k \neq 6$. Let F and G be (Q, Q) -biprojective functions over $\mathbb{F}_q$. Assume $k \nmid 2\ell$ and $r \nmid [C_F : Z]$. Then F, G are EL-equivalent if and only if they are linear equivalent via a mapping $\gamma = \text{diag}(M, L) \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2})$, where, by writing $M = \begin{pmatrix} M_1 & M_2 \\ M_3 & M_4 \end{pmatrix}$ and $L = \begin{pmatrix} L_1 & L_2 \\ L_3 & L_4 \end{pmatrix}$, the subfunctions $M_i, L_i : \mathbb{F}_q \rightarrow \mathbb{F}_q$ are monomials of the same degree, say 2^t for some t , or zero.

Remark 4 The linear permutations M and L in Theorem 16 can be explicitly written as

$$M(x, y) = (\beta_1 x^{2^t} + \beta_2 y^{2^t}, \beta_3 x^{2^t} + \beta_4 y^{2^t}) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y]$$

and

$$L(x, y) = (\delta_1 x^{2^t} + \delta_2 y^{2^t}, \delta_3 x^{2^t} + \delta_4 y^{2^t}) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y]$$

for some $\beta_i, \delta_i \in \mathbb{F}_q$ with $\beta_1\beta_4 \neq \beta_2\beta_3, \delta_1\delta_4 \neq \delta_2\delta_3$ and some non-negative integer t . In other words, F and G are EL- (or EA-) equivalent if and only if they are linear equivalent via a composition of an $\mathcal{L}$ -equivalence with $(\cdot)^{(2^t)}$ for some non-negative integer t . Since linear equivalence implies EL-equivalence, this proves the converse of Remark 3 for (Q, Q) -biprojective functions (and DO polynomials in the form (6)) as long as $k > 2, k \neq 6, k \nmid 2\ell$ and $r \nmid [C_F : Z]$.

Remark 5 We further note that a similar statement to Theorem 16 holds true for $q = p^k$ and $Q = p^\ell$ with p being an odd prime and $k > 2$ (note $k \neq 6$ is no longer necessary since it is only assumed when $p = 2$ in order that the primitive prime divisor r exists). However for the purpose of this paper we write the statement for $p = 2$ only.

Before giving the proof of Theorem 16, we need a few preliminary results.

Lemma 17 [13, Lemma 4] Let $k > 2, k \neq 6$ and F be a (Q, Q) -biprojective function, such that C_F contains Z as a subgroup with index I and r does not divide I . Then Z is a Sylow r -subgroup of $\text{Aut}_{\text{EL}}(F)$.

Remark 6 [13, Lemma 4] was stated for (Q, Q') -biprojective functions with Q, Q' being general powers of 2 that are not necessarily congruent modulo $q - 1$. For brevity and purpose of our paper we state it for $Q' = Q$ only.

Lemma 18 ([14, Lemma 5.7], [13, Lemma 5]) Denote Λ to be the collection of all linear mappings on $\mathbb{F}_q \times \mathbb{F}_q$,

$$N_{\mathcal{R}} := \{\gamma \in \Lambda : \Delta_{\mathcal{R}}\gamma = \gamma\Delta_{\mathcal{R}}\}$$

and

$$C_{\mathcal{R}} := \{\gamma \in \Lambda : \delta\gamma = \gamma\delta \text{ for all } \delta \in \Delta_{\mathcal{R}}\}.$$

Then

$$\begin{aligned} 1. \quad N_{\mathcal{R}} &= \left\{ \begin{pmatrix} m_{c_1}\tau & m_{c_2}\tau \\ m_{c_3}\tau & m_{c_4}\tau \end{pmatrix} : c_1, c_2, c_3, c_4 \in \mathbb{F}_q, \tau \in \text{Gal}(\mathbb{F}_q/\mathbb{F}_2) \right\}; \\ 2. \quad C_{\mathcal{R}} &= \left\{ \begin{pmatrix} m_{c_1} & m_{c_2} \\ m_{c_3} & m_{c_4} \end{pmatrix} : c_1, c_2, c_3, c_4 \in \mathbb{F}_q \right\}. \end{aligned}$$

Remark 7 [14, Lemma 5.7] and [13, Lemma 5] were stated as normalizers and centralizers of $\Delta_{\mathcal{R}}$, which are restricted to invertible linear mappings. However from their proof it is easy to see that it works for non-invertible ones as well. Due to usage for some of the arguments later, we state this general version instead.

Proof of Theorem 16 Assume F, G are EL-equivalent by the EL-mapping $\alpha \in \text{ELM}$, that is, $\alpha(\Gamma_F) = \Gamma_G$. Then $\alpha^{-1}Z_{\mathcal{R}}\alpha$ is a subgroup of $\text{Aut}_{\text{EL}}(F)$ of size $\#Z_{\mathcal{R}} = R$. Since $r \nmid [C_F : Z]$, $\alpha^{-1}Z_{\mathcal{R}}\alpha$ is a Sylow r -subgroup of $\text{Aut}_{\text{EL}}(F)$ by Lemma 17. By Sylow's Theorem, there exists $\lambda \in \text{Aut}_{\text{EL}}(F)$ such that $\gamma := \alpha\lambda \in \text{ELM}$ satisfies

$$\gamma^{-1}Z_{\mathcal{R}}\gamma = \lambda^{-1}(\alpha^{-1}Z_{\mathcal{R}}\alpha)\lambda = Z_{\mathcal{R}},$$

that is, $\gamma \in N_{\text{ELM}}(Z_{\mathcal{R}})$. It is easy to see that γ satisfies $\gamma(\Gamma_F) = \Gamma_G$ as well. Write

$$\gamma = \begin{pmatrix} M & 0 \\ N & L \end{pmatrix}$$

with $M, L \in \text{GL}(\mathbb{F}_{q^2})$, which implies $G \circ M = L \circ F + N$.

Now for any $a \in \mathcal{R}$, we have

$$\gamma^{-1} \text{diag}(m_a, m_a, m_{a^{Q+1}}, m_{a^{Q+1}}) \gamma = \begin{pmatrix} M^{-1} \text{diag}(m_a, m_a) M & 0 \\ * & L^{-1} \text{diag}(m_{a^{Q+1}}, m_{a^{Q+1}}) L \end{pmatrix} \in Z_{\mathcal{R}},$$

that is,

$$M^{-1} \text{diag}(m_a, m_a) M = \text{diag}(m_b, m_b), \quad L^{-1} \text{diag}(m_{a^{Q+1}}, m_{a^{Q+1}}) L = \text{diag}(m_{b^{Q+1}}, m_{b^{Q+1}}) \quad (13)$$

for some $b = \pi(a)$, where $\pi: \mathcal{R} \rightarrow \mathcal{R}$ is a permutation.

We now claim that $a \mapsto a^{Q+1}$ permutes $\mathcal{R}$. Recall that R is a power of r . Hence to prove the claim, it suffices to show that $r \nmid (Q+1)$.

Denote $e := \gcd(k, \ell)$. If k/e is odd then Lemma 3 implies $\gcd(Q+1, q-1) = 1$ and $r \mid (q-1)$, so $r \nmid (Q+1)$ immediately follows.

We now assume k/e is even. Then Lemma 3 yields $\gcd(Q+1, q-1) = 2^e + 1$. If $r \mid (Q+1)$, then $r \mid (2^e + 1)$ and so $r \mid (2^{2e} - 1)$. By the definition of r , we must have $k \mid 2e$. Since k/e is even, the only choice is $k = 2e$. But then this means $\frac{k}{2} \mid \ell$ and so $k \mid 2\ell$, a contradiction.

By (13) and the claim, then M and L are both in the normalizer of $\Delta_{\mathcal{R}}$. Hence by Lemma 18, we have that M_i 's are zero or monomials of the same degree, say 2^t , and L_i 's are zero or monomials of the same degree, say 2^s .

Writing $G(x, y) = (g_1(x, y), g_2(x, y)) \in \mathbb{F}_q[x, y] \times \mathbb{F}_q[x, y]$ where

$$g_i(x, y) = a_i x^{Q+1} + b_i x^Q y + c_i x y^Q + d_i y^{Q+1}, i = 1, 2,$$

and $M_j(x) := \beta_j x^{2^t}$ for $j = 1, 2, 3, 4$, we obtain

$$(G \circ M)(x, y) = (g_1(M_1(x) + M_2(y), M_3(x) + M_4(y)), g_2(M_1(x) + M_2(y), M_3(x) + M_4(y))),$$

with

$$g_i((M_1(x) + M_2(y), M_3(x) + M_4(y))) = a_i(\beta_1 x^{2^t} + \beta_2 y^{2^t})^{Q+1} + b_i(\beta_1 x^{2^t} + \beta_2 y^{2^t})^Q (\beta_3 x^{2^t} + \beta_4 y^{2^t}) \\ + c_i(\beta_1 x^{2^t} + \beta_2 y^{2^t})(\beta_3 x^{2^t} + \beta_4 y^{2^t})^Q + d_i(\beta_3 x^{2^t} + \beta_4 y^{2^t})^{Q+1}$$

for $i = 1, 2$. We now see that the only terms appearing in $G \circ M$ are of the form $x^{2^t(Q+1)}, x^{2^t Q} y^{2^t}, x^{2^t} y^{2^t Q}$ or $y^{2^t(Q+1)}$, and those for $L \circ F$ are of the form $x^{2^s(Q+1)}, x^{2^s Q} y^{2^s}, x^{2^s} y^{2^s Q}$ or $y^{2^s(Q+1)}$. Since $k \nmid \ell$, $Q + 1$ is not (congruent modulo $q - 1$ to) a power of two and hence neither $G \circ M$ nor $L \circ F$ contain linear terms. Therefore $N = 0$ and $G = L \circ F \circ M^{-1}$ is linear equivalent to F . Now $G \circ M = L \circ F$ is possible only if $2^s(Q + 1) \equiv 2^t(Q + 1) \pmod{q - 1}$ and $\{2^s Q, 2^s\} \equiv \{2^t Q, 2^t\} \pmod{q - 1}$. Note that here the former is implied by the latter, and so we only need to consider the latter. It means either of the following hold:

1. $2^s Q \equiv 2^t Q \pmod{q - 1}$ and $2^s \equiv 2^t \pmod{q - 1}$ or
2. $2^s Q \equiv 2^t \pmod{q - 1}$ and $2^s \equiv 2^t Q \pmod{q - 1}$.

Assume 2. above holds. Combining both congruences yields

$$2^t Q^2 \equiv 2^t \pmod{q - 1},$$

which further implies

$$Q^2 \equiv 1 \pmod{q - 1}.$$

But then this means $k \mid 2\ell$, which is a contradiction with the assumption.

Hence it must be 1. that holds, which implies $s \equiv t \pmod{k}$, and we may simply take $s = t$ since L is defined over $\mathbb{F}_q \times \mathbb{F}_q$.

This completes the proof of Theorem 16. □

Lemma 19 *Assume $k > 2, k \neq 6$ and $k \nmid 2\ell$. Let F be defined by the following (Q, Q) -biprojective function over $\mathbb{F}_q \times \mathbb{F}_q$ or polynomial over $\mathbb{F}_{q^2}$:*

1. $F(x, y) = (x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1})$ with $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2$;
2. $F(X) = X^{Q+1}$;

3. $F(X) = X^{Q+q} + \varepsilon X^{Q+1}$ with $\varepsilon \in \mathbb{F}_q \setminus \{1\}$.

Then $r \nmid [C_F : Z]$.

Proof Let $\lambda = \begin{pmatrix} M & 0 \\ N & L \end{pmatrix} \in C_F$. Then for any $a \in \mathcal{R}$, we have

$$\begin{aligned} \lambda^{-1} \text{diag}(m_a, m_a, m_{a^{Q+1}}, m_{a^{Q+1}}) \lambda &= \begin{pmatrix} M^{-1} \text{diag}(m_a, m_a) M & 0 \\ * & L^{-1} \text{diag}(m_{a^{Q+1}}, m_{a^{Q+1}}) L \end{pmatrix} \\ &= \begin{pmatrix} \text{diag}(m_a, m_a) & 0 \\ 0 & \text{diag}(m_{a^{Q+1}}, m_{a^{Q+1}}) \end{pmatrix}. \end{aligned} \quad (14)$$

Similar to the proof of Theorem 16, we now see that M and L are both in the centralizer of $\Delta_{\mathcal{R}}$. Hence by Lemma 18, there are $c_i, d_i \in \mathbb{F}_q$ for $i = 1, 2, 3, 4$ such that

$$M = \begin{pmatrix} m_{c_1} & m_{c_2} \\ m_{c_3} & m_{c_4} \end{pmatrix}$$

and

$$L = \begin{pmatrix} m_{d_1} & m_{d_2} \\ m_{d_3} & m_{d_4} \end{pmatrix}$$

respectively. Note here since $M, L \in \text{GL}(\mathbb{F}_{q^2})$, we must have $c_1 c_4 \neq c_2 c_3$ and $d_1 d_4 \neq d_2 d_3$.

Since $\lambda \in \text{Aut}_{\text{EL}}(F)$, we have $F \circ M = L \circ F + N$. As F is (equivalent to) a (Q, Q) -biprojective function, similar to the proof of Theorem 16, we can see that $N = 0$ as long as $k \nmid \ell$.

1. We have

$$\begin{aligned} &F \circ M(x, y) \\ &= F(c_1 x + c_2 y, c_3 x + c_4 y) \\ &= ((c_1 x + c_2 y)^{Q+1} + (c_1 x + c_2 y)^Q (c_3 x + c_4 y), (c_1 x + c_2 y)(c_3 x + c_4 y)^Q + \varepsilon (c_3 x + c_4 y)^{Q+1}) \\ &= ((c_1^Q (c_1 + c_3), c_1^Q (c_2 + c_4), c_2^Q (c_1 + c_3), c_2^Q (c_2 + c_4))_Q, (c_3^Q (c_1 + \varepsilon c_3), c_3^Q (c_2 + \varepsilon c_4), \end{aligned}$$

$$c_4^Q(c_1 + \varepsilon c_3), c_4^Q(c_2 + \varepsilon c_4))_Q)$$

and

$$\begin{aligned} L \circ F(x, y) &= (d_1(1, 1, 0, 0)_Q + d_2(0, 0, 1, \varepsilon)_Q, d_3(1, 1, 0, 0)_Q + d_4(0, 0, 1, \varepsilon)_Q) \\ &= ((d_1, d_1, d_2, \varepsilon d_2)_Q, (d_3, d_3, d_4, \varepsilon d_4)_Q). \end{aligned}$$

This means we are solving the system of 8 equations given by

$$c_1^Q(c_1 + c_3) = d_1 \tag{15}$$

$$c_1^Q(c_2 + c_4) = d_1 \tag{16}$$

$$c_2^Q(c_1 + c_3) = d_2 \tag{17}$$

$$c_2^Q(c_2 + c_4) = \varepsilon d_2 \tag{18}$$

$$c_3^Q(c_1 + \varepsilon c_3) = d_3 \tag{19}$$

$$c_3^Q(c_2 + \varepsilon c_4) = d_3 \tag{20}$$

$$c_4^Q(c_1 + \varepsilon c_3) = d_4 \tag{21}$$

$$c_4^Q(c_2 + \varepsilon c_4) = \varepsilon d_4. \tag{22}$$

Adding (15) and (16) gives

$$c_1^Q(c_1 + c_2 + c_3 + c_4) = 0,$$

so that either of the following holds:

$$(A1) \quad c_1 = 0;$$

$$(A2) \quad c_1 + c_2 + c_3 + c_4 = 0.$$

Similarly, adding (19) and (20) gives

$$c_3^Q[c_1 + c_2 + \varepsilon(c_3 + c_4)] = 0,$$

so that either of the following holds:

(B1) $c_3 = 0$;

(B2) $c_1 + c_2 + \varepsilon(c_3 + c_4) = 0$.

Since $c_1c_4 \neq c_2c_3$, we know that (A1) and (B1) cannot hold simultaneously. Moreover, if (A2) and (B2) hold simultaneously, then adding them yields $(1+\varepsilon)(c_3+c_4) = 0$, and since $\varepsilon \neq 1$, we must have $c_3 = c_4$, which further implies $c_1 = c_2$ by (A2) (or (B2)) too, and hence $c_1c_4 = c_2c_3$ which is again a contradiction.

Hence it must be either ((A2) and (B1)) or ((A1) and (B2)) (but not both) hold.

Case 1: (A2) and (B1) hold.

Adding (17) and (18) gives

$$0 = c_2^Q(c_1 + c_2 + c_3 + c_4) = (1 + \varepsilon)d_2,$$

and so $d_2 = 0$ since $\varepsilon \neq 1$. If $c_2 \neq 0$, then (17) and (18) yield $c_1 = c_3$ and $c_2 = c_4$, whence we again arrive at the contradiction $c_1c_4 = c_2c_3$. Hence $c_2 = 0$. Moreover, by (19) (or (20)), we have $d_3 = 0$. (21) and (22) give $c_4^Qc_1 = d_4$ and $\varepsilon c_4^{Q+1} = \varepsilon d_4$. Since $0 = c_2c_3 \neq c_1c_4$ and $0 = d_2d_3 \neq d_1d_4$, we see that $c_4 = c_1$. Finally by (15) and (21), we conclude that $d_4 = d_1 = c_1^{Q+1}$. It can be seen that the remaining equations are satisfied automatically. This case hence yields Z precisely.

Case 2: (A1) and (B2) hold.

Adding (21) and (22) gives

$$0 = c_4^Q(c_1 + c_2 + \varepsilon(c_3 + c_4)) = (1 + \varepsilon)d_4,$$

and so $d_4 = 0$ since $\varepsilon \neq 1$. If $c_4 \neq 0$, then (21) and (22) yield $c_1 = \varepsilon c_3$ and $c_2 = \varepsilon c_4$, whence we again arrive at the contradiction $c_1 c_4 = c_2 c_3$. Hence $c_4 = 0$. Moreover, by (15) (or (16)), we have $d_1 = 0$. (17) and (18) give $c_2^Q c_3 = d_2$ and $c_2^{Q+1} = \varepsilon d_2$. Since $0 = c_1 c_4 \neq c_2 c_3$ and $0 = d_1 d_4 \neq d_2 d_3$, we see that $c_2 = \varepsilon c_3$. Finally by (17) and (19), we conclude that $d_2 = \varepsilon^Q c_3^{Q+1}$ and $d_3 = \varepsilon c_3^{Q+1}$. It can be seen that the remaining equations are satisfied automatically. This case hence yields the set

$$\left\{ \begin{pmatrix} 0 & m_{\varepsilon a} & 0 & 0 \\ m_a & 0 & 0 & 0 \\ 0 & 0 & 0 & m_{\varepsilon^Q a^{Q+1}} \\ 0 & 0 & m_{\varepsilon a^{Q+1}} & 0 \end{pmatrix} : a \in \mathbb{F}_q^* \right\}.$$

Combining both, then we easily see that $\sharp C_F = 2(q-1)$, or equivalently $[C_F : Z] =$

2. Since r is an odd prime, $r \nmid [C_F : Z]$.

2. Using the linear bijection T as defined in (5), the linear mappings M and L may be rewritten as polynomials over $\mathbb{F}_{q^2}$ given by $M(X) = \alpha \bar{X} + \beta X$ and $L(X) = \gamma \bar{X} + \delta X$, where $\bar{X} := X^q$, $\alpha, \beta, \gamma, \delta \in \mathbb{F}_{q^2}$ are such that $\alpha^{q+1} \neq \beta^{q+1}$ and $\gamma^{q+1} \neq \delta^{q+1}$.

Therefore we obtain

$$\begin{aligned} F \circ M(X) &= (\alpha \bar{X} + \beta X)^{Q+1} \\ &= \alpha^{Q+1} \bar{X}^{Q+1} + \alpha^Q \beta \bar{X}^Q X + \alpha \beta^Q \bar{X} X^Q + \beta^{Q+1} X^{Q+1} \end{aligned}$$

and

$$L \circ F(X) = \gamma \bar{X}^{Q+1} + \delta X^{Q+1}$$

respectively.

This means we are solving the system of 4 equations

$$\alpha^{Q+1} = \gamma \quad (23)$$

$$\alpha^Q \beta = 0 \quad (24)$$

$$\alpha \beta^Q = 0 \quad (25)$$

$$\beta^{Q+1} = \delta. \quad (26)$$

From (24) (or 25) we easily see that either $\alpha = 0$ or $\beta = 0$. If $\alpha = 0$, then (23) yields $\gamma = 0$, and since $\delta^{q+1} \neq \gamma^{q+1}$, we see that $\delta \neq 0$ and $\beta \neq 0$ too. (26) gives $q^2 - 1$ possibilities. Similarly, if $\beta = 0$, we get $\delta = 0$ by (26), and we get another $q^2 - 1$ possibilities by (23). Here we note that Z corresponds to the case $\alpha = 0$ and $\beta \in \mathbb{F}_q^*$. Combining all, we get $\sharp C_F = 2(q^2 - 1)$, so that $[C_F : Z] = 2(q + 1)$. We know that r is odd. If $r \mid [C_F : Z]$, then $r \mid (q + 1)$. But we know that $r \mid (q - 1)$. Hence r divides $(q + 1) - (q - 1) = 2$ which is a contradiction.

3. Using the linear bijection T as defined in (5), the linear mappings M and L may be rewritten as polynomials over $\mathbb{F}_{q^2}$ given by $M(X) = \alpha \bar{X} + \beta X$ and $L(X) = \gamma \bar{X} + \delta X$, where $\bar{X} := X^q$, $\alpha, \beta, \gamma, \delta \in \mathbb{F}_{q^2}$ are such that $\alpha^{q+1} \neq \beta^{q+1}$ and $\gamma^{q+1} \neq \delta^{q+1}$.

Therefore we obtain

$$\begin{aligned} F \circ M(X) &= (\bar{\beta} \bar{X} + \bar{\alpha} X)(\alpha \bar{X} + \beta X)^Q + \varepsilon(\alpha \bar{X} + \beta X)^{Q+1} \\ &= \alpha^Q(\bar{\beta} + \varepsilon \alpha) \bar{X}^{Q+1} + \alpha^Q(\bar{\alpha} + \varepsilon \beta) \bar{X}^Q X + \beta^Q(\bar{\beta} + \varepsilon \alpha) \bar{X} X^Q + \beta^Q(\bar{\alpha} + \beta) X^{Q+1} \end{aligned}$$

and

$$L \circ F(X) = \varepsilon \gamma \bar{X}^{Q+1} + \gamma \bar{X}^Q X + \delta \bar{X} X^Q + \varepsilon \delta X^{Q+1}$$

respectively.

This means we are solving the system of 4 equations

$$\alpha^Q(\bar{\beta} + \varepsilon\alpha) = \varepsilon\gamma \quad (27)$$

$$\alpha^Q(\bar{\alpha} + \varepsilon\beta) = \gamma \quad (28)$$

$$\beta^Q(\bar{\beta} + \varepsilon\alpha) = \delta \quad (29)$$

$$\beta^Q(\bar{\alpha} + \varepsilon\beta) = \varepsilon\delta. \quad (30)$$

Assuming $\gamma\delta \neq 0$, so we have $\alpha\beta(\bar{\beta} + \varepsilon\alpha)(\bar{\alpha} + \varepsilon\beta) \neq 0$. In particular $\varepsilon \neq 0$ too.

Dividing (28) by (27) yields

$$\frac{\bar{\alpha} + \varepsilon\beta}{\bar{\beta} + \varepsilon\alpha} = \varepsilon^{-1}.$$

Dividing (30) by (29) yields

$$\frac{\bar{\alpha} + \varepsilon\beta}{\bar{\beta} + \varepsilon\alpha} = \varepsilon.$$

Hence $\varepsilon^{-1} = \varepsilon$, implying $\varepsilon = 1$ which is a contradiction.

Therefore either $\gamma = 0$ or $\delta = 0$.

Case 1: $\gamma = 0$.

(27) and (28) imply that either of the following holds:

(C1) $\alpha = 0$;

(C2) $\bar{\beta} + \varepsilon\alpha = 0 = \bar{\alpha} + \varepsilon\beta$.

We first assume (C1) holds. Then (29) and (30) give $\beta^Q\bar{\beta} = \delta$ and $\varepsilon\beta^{Q+1} = \varepsilon\delta$.

Since $0 = \alpha^{Q+1} \neq \beta^{Q+1}$ and $0 = \gamma^{q+1} \neq \delta^{q+1}$, we see that either $\varepsilon = 0$ or $\beta \in \mathbb{F}_q^*$.

In the latter case, by (29) (or (30)), we have $\delta = \beta^{Q+1} \in \mathbb{F}_q^*$ too.

Now assume instead (C2) holds. Then we have $\overline{\alpha + \beta} = \varepsilon(\alpha + \beta)$, so that either $\alpha = \beta$ or $\varepsilon = (\alpha + \beta)^{q-1} \in \mu_{q+1}$. Since $\varepsilon \in \mathbb{F}_q$, the latter implies $\varepsilon = 1$ which is a contradiction. Hence $\alpha = \beta$ and this clearly contradicts with $\alpha^{q+1} \neq \beta^{q+1}$.

Overall, we have $M(X) = \beta X$ and $L(X) = \beta^Q\bar{\beta}X$ for some $\beta \in \mathcal{F}$, where $\mathcal{F} := \mathbb{F}_{q^2}^*$ if $\varepsilon = 0$ and $\mathcal{F} := \mathbb{F}_q^*$ otherwise. The number of (M, L) -pairs corresponding

to this case is hence $q^2 - 1$ if $\varepsilon = 0$ and $q - 1$ otherwise. Note that the subcase $\beta \in \mathbb{F}_q^*$ corresponds to Z precisely.

Case 2: $\delta = 0$.

Due to the symmetry of the equation, we can simply mimic the arguments in the previous case to conclude that $\beta = 0$, either $\varepsilon = 0$ or $\alpha \in \mathbb{F}_q^*$, and $\gamma = \alpha^Q \bar{\alpha}$.

Hence in this case we have $M(X) = \alpha \bar{X}$ and $L(X) = \alpha^Q \bar{\alpha} \bar{X}$ for $\alpha \in \mathcal{F}$. The number of (M, L) -pairs corresponding to this case is then again $q^2 - 1$ if $\varepsilon = 0$ and $q - 1$ otherwise.

Combining both cases, we see that $\sharp C_F = 2(q^2 - 1)$ if $\varepsilon = 0$ and $\sharp C_F = 2(q - 1)$ otherwise, or equivalently, $[C_F : Z] = 2(q + 1)$ if $\varepsilon = 0$ and $[C_F : Z] = 2$ otherwise. It is clear that $r \nmid [C_F : Z]$ in the latter case. The former case follows from a similar argument as Class 2.

This completes the proof of Lemma 19. $\square$

Theorem 20 *Assume $k > 2$, $k \neq 6$ and $k \nmid 2\ell$. Let θ be a fixed primitive element of $\mathbb{F}_q^*$. The EA- (resp. EL-) equivalence class representatives of the collection of functions*

$$\mathcal{P} := \{(x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1}) : \varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2\} \cup \{X^{Q+q} + \varepsilon X^{Q+1} : \varepsilon \in \mathbb{F}_q \setminus \{1\}\} \cup \{X^{Q+1}\}$$

are given by:

1. X^{Q+1} ;
2. X^{Q+q} ;
3. $(x^{Q+1} + x^Q y, xy^Q + \theta^i y^{Q+1})$, where i varies over $\mathcal{C}$;
4. $X^{Q+q} + \theta^i X^{Q+1}$, where i varies over $\mathcal{C}$.

Proof Let $f_1, f_2 \in \mathcal{P}$. By Lemma 19 and Remark 4, f_1 and f_2 are EA- (or EL-) equivalent if and only if they are linear equivalent via a composition of $\mathcal{L}$ -equivalence and $(\cdot)^{(2^t)}$ for some non-negative integer t . Then the desired result follows from Theorem 15. $\square$

Finally, combining Theorems 13 and 20 completes the proof of Statement 2(a) of Theorem 2.

4.3 CCZ-equivalence

In this section we complete the proof of Statement 2(b) of Theorem 2 by studying the most general equivalence, namely CCZ-equivalence, among the functions listed in Theorem 15.

First, we prove that the result in [20, Proposition 2.2] can be generalized to CCZ-equivalence.

Proposition 21 *Let f, g be quadratic functions over $\mathbb{F}_q$ which fix 0. If f, g are CCZ-equivalent, then there is $\gamma \in \text{GL}(\mathbb{F}_q \times \mathbb{F}_q)$ such that $\gamma(\Gamma_f) = \Gamma_g$. In other words, the affine permutation between the corresponding graphs can be assumed to be linear.*

Proof Since f, g are CCZ-equivalent, there are linear functions $M, K, N, L: \mathbb{F}_q \times \mathbb{F}_q \rightarrow \mathbb{F}_q \times \mathbb{F}_q$ and $\alpha, \beta \in \mathbb{F}_q \times \mathbb{F}_q$ such that

$$\gamma' := \begin{pmatrix} M & K \\ N & L \end{pmatrix} \in \text{GL}(\mathbb{F}_q \times \mathbb{F}_q)$$

and

$$g \circ (M + K \circ f + \alpha) = N + L \circ f + \beta.$$

Define the function $D_{M,K,f,g,\alpha}$ by

$$D_{M,K,f,g,\alpha} = g \circ (M + K \circ f + \alpha) + g \circ (M + K \circ f) + g(\alpha).$$

Evaluating at 0, we easily see that $D_{M,K,f,g,\alpha}(0) = 0$.

Combining both, we get

$$g \circ (M + K \circ f) = D_{M,K,f,g,\alpha} + N + L \circ f + \beta + g(\alpha).$$

Evaluating at 0 again, noting that $f(0) = g(0) = 0$, we have

$$0 = g \circ (M + K \circ f)(0) = \beta + g(\alpha),$$

that is, $g(\alpha) = \beta$.

Hence

$$g \circ (M + K \circ f) = D_{M,K,f,g,\alpha} + N + L \circ f.$$

We now show that there are linear mappings $L_0, N_0 : \mathbb{F}_q \rightarrow \mathbb{F}_q$ such that

$$D_{M,K,f,g,\alpha} = N_0 + L_0 \circ f,$$

so that the desired statement holds for

$$\gamma = \begin{pmatrix} M & K \\ N + N_0 & L + L_0 \end{pmatrix}.$$

Since g is quadratic and fixes 0, it can be written in the form $g(x) = \sum_{0 \leq i < j \leq k-1} a_{ij} x^{2^i + 2^j} + A(x)$ where $a_{ij} \in \mathbb{F}_q$ and A is linear (the summation term is essentially just a DO polynomial over $\mathbb{F}_q$). Then

$$\begin{aligned} D_{M,K,f,g,\alpha}(x) &= \sum_{0 \leq i < j \leq k-1} a_{ij} \{ [M(x) + K(f(x)) + \alpha]^{2^i + 2^j} + [M(x) + K(f(x))]^{2^i + 2^j} + \alpha^{2^i + 2^j} \} \\ &= \sum_{0 \leq i < j \leq k-1} a_{ij} \{ \alpha^{2^i} [M(x) + K(f(x))]^{2^j} + \alpha^{2^j} [M(x) + K(f(x))]^{2^i} \}, \end{aligned}$$

so we can take

$$N_0 = \mathcal{L} \circ M, \quad L_0 = \mathcal{L} \circ K$$

where $\mathcal{L} : \mathbb{F}_q \rightarrow \mathbb{F}_q$ is the linear polynomial

$$\mathcal{L}(x) = \sum_{0 \leq i < j \leq k-1} a_{ij} (\alpha^{2^i} x^{2^j} + \alpha^{2^j} x^{2^i}).$$

Finally, we have

$$\gamma = \begin{pmatrix} m_1 & 0 \\ \mathcal{L} & m_1 \end{pmatrix} \gamma' \in \text{GL}(\mathbb{F}_q \times \mathbb{F}_q)$$

□

as desired.

Theorem 22 Assume $k > 2$ and $k \notin \{4, 6, 10, 12, 18\}$. Let F and G be (Q, Q) -biprojective functions over $\mathbb{F}_q \times \mathbb{F}_q$ that are not both permutations. Assume $(Q + 1)^k \not\equiv 0, 1 \pmod{R}$ and $r \nmid [C'_F : Z]$. Then F, G are CCZ-equivalent if and only if they are linear equivalent via a mapping $\gamma = \text{diag}(M, L) \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2})$, where, by writing $M = \begin{pmatrix} M_1 & M_2 \\ M_3 & M_4 \end{pmatrix}$ and $L = \begin{pmatrix} L_1 & L_2 \\ L_3 & L_4 \end{pmatrix}$, the subfunctions $M_i, L_i: \mathbb{F}_q \rightarrow \mathbb{F}_q$ are monomials of the same degree, say 2^t for some t , or zero.

- Remark 8* 1. The assumptions of Theorem 22 immediately imply $k \nmid 2\ell$: indeed if $k \mid 2\ell$, then $Q^2 \equiv 1 \pmod{q-1}$ and so $Q^2 \equiv 1 \pmod{R}$ or equivalently $Q \equiv \pm 1 \pmod{R}$ (since R is an odd prime power). Then $(Q + 1)^k \equiv 0, 2^k \equiv 0, 1 \pmod{R}$.
2. A prime r is called a *large Zsigmondy prime* for $(2, k)$ if it is a primitive prime divisor of $2^k - 1$ such that either $r > k + 1$ or $r^2 \mid (2^k - 1)$ (equivalently, $R > k + 1$). Feit [11] shows that large Zsigmondy primes for $(2, k)$ exist except when $k \in \{1, 4, 6, 10, 12, 18\}$. Excluding the cases $k \in \{1, 6\}$ where primitive prime divisors do not exist at all, in other exceptional cases ($k \in \{4, 10, 12, 18\}$) the only primitive prime divisor r is such that 2 is primitive modulo r and $R = r$. Then we either have $(Q + 1)^k \equiv 0 \pmod{R}$ or $(Q + 1)^k \equiv 2^{km} \equiv 1 \pmod{R}$ for some integer m . Thereby we exclude those k 's in Theorem 22.
3. Theorem 22 is no longer valid if we consider (Q, Q') -biprojective functions with Q' and Q being distinct powers of 2 $\pmod{q-1}$.

Before proving Theorem 22, we need the following modified result from Lemma 17, by simply noting that its proof (as stated in [13, Lemma 4]) remains valid with C_F and $\text{Aut}_{\text{EL}}(F)$ replaced by C'_F and $\text{Aut}(F)$ respectively:

Lemma 23 Let $k > 2, k \neq 6$ and F be a (Q, Q) -biprojective function, such that C'_F contains Z as a subgroup with index I and r does not divide I . Then Z is a Sylow r -subgroup of $\text{Aut}(F)$.

Proof of Theorem 22 Assume F, G are CCZ-equivalent. By Proposition 21, there is $\alpha \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2})$, such that $\alpha(\Gamma_F) = \Gamma_G$. Then $\alpha^{-1}Z_{\mathcal{R}}\alpha$ is a subgroup of $\text{Aut}(F)$ of size $\sharp Z_{\mathcal{R}} = R$. Since $r \nmid [C'_F : Z]$, $\alpha^{-1}Z_{\mathcal{R}}\alpha$ is a Sylow r -subgroup of $\text{Aut}(F)$ by Lemma 23. By Sylow's Theorem, there exists $\lambda \in \text{Aut}(F)$ such that $\gamma := \alpha\lambda \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2})$ satisfies

$$\gamma^{-1}Z_{\mathcal{R}}\gamma = \lambda^{-1}(\alpha^{-1}Z_{\mathcal{R}}\alpha)\lambda = Z_{\mathcal{R}},$$

that is, $\gamma \in N_{\text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2})}(Z_{\mathcal{R}})$. It is easy to see that γ satisfies $\gamma(\Gamma_F) = \Gamma_G$ as well. Write

$$\gamma = \begin{pmatrix} M & K \\ N & L \end{pmatrix},$$

which implies $G \circ (M + K \circ F) = N + L \circ F$.

Now for any $a \in \mathcal{R}$, we have

$$\text{diag}(m_a, m_a, m_{a^{Q+1}}, m_{a^{Q+1}})\gamma = \gamma \text{diag}(m_b, m_b, m_{b^{Q+1}}, m_{b^{Q+1}})$$

for some $b = \pi(a)$, where $\pi: \mathcal{R} \rightarrow \mathcal{R}$ is a permutation.

This implies

$$\begin{cases} \text{diag}(m_a, m_a)M &= M \text{diag}(m_b, m_b), \\ \text{diag}(m_a, m_a)K &= K \text{diag}(m_{b^{Q+1}}, m_{b^{Q+1}}), \\ \text{diag}(m_{a^{Q+1}}, m_{a^{Q+1}})N &= N \text{diag}(m_b, m_b), \\ \text{diag}(m_{a^{Q+1}}, m_{a^{Q+1}})L &= L \text{diag}(m_{b^{Q+1}}, m_{b^{Q+1}}). \end{cases} \quad (31)$$

Since $k \nmid 2\ell$, as in the proof of Theorem 16, x^{Q+1} permutes $\mathcal{R}$. Hence M, K, N, L satisfy $\Delta_{\mathcal{R}}M = M\Delta_{\mathcal{R}}, \Delta_{\mathcal{R}}K = K\Delta_{\mathcal{R}}, \Delta_{\mathcal{R}}N = N\Delta_{\mathcal{R}}$ and $\Delta_{\mathcal{R}}L = L\Delta_{\mathcal{R}}$. By Lemma 18, M_i 's, K_i 's, N_i 's and L_i 's are zero or monomials of degrees say $2^t, 2^s, 2^u$ and 2^v respectively. In fact by looking through the proof (in [14, Lemma 5.7]), we see that t, s, u, v must satisfy

$$a = b^{2^t}, a = b^{2^s(Q+1)}, a^{Q+1} = b^{2^u}, a^{Q+1} = b^{2^v(Q+1)}$$

for any $a \in \mathcal{R}$, if M, K, N, L is nonzero, respectively.

Now it is easy to see that if M and L are both nonzero, then $v = t$. Moreover, if M and K are both nonzero, then we have $2^t \equiv 2^s(Q+1) \pmod{R}$, or equivalently, $2^{t-s} \equiv Q+1$

(mod R). But then this implies $(Q+1)^k \equiv q^{t-s} \equiv 1 \pmod{R}$, which contradicts with the given assumption. Similar contradiction occurs if any of the following pairs are both nonzero: $(K, L), (M, N), (N, L)$. Hence either $M = L = 0$ or $K = N = 0$.

Assume $M = L = 0$. Then $K, N \in \text{GL}(\mathbb{F}_{q^2})$ and we have $G \circ K \circ F = N \in \text{GL}(\mathbb{F}_{q^2})$. That is, $G \circ K \circ F$ is bijective. This implies that G is surjective and F is injective. But since the domains and codomains of F and G are all $\mathbb{F}_q \times \mathbb{F}_q$, F and G are both permutations over $\mathbb{F}_q \times \mathbb{F}_q$, which is a contradiction.

Hence it must be the case that $K = N = 0$, and so $\gamma = \text{diag}(M, L) \in \text{GL}(\mathbb{F}_{q^2} \times \mathbb{F}_{q^2})$, where M_i 's and L_i 's are either zero or monomials of the same degree 2^t .

This completes the proof of Theorem 22. $\square$

Lemma 24 *Assume $k > 2, k \notin \{4, 6, 10, 12, 18\}$ and $(Q+1)^k \not\equiv 0, 1 \pmod{R}$. Let F be defined by the following (Q, Q) -biprojective function over $\mathbb{F}_q \times \mathbb{F}_q$ or polynomial over $\mathbb{F}_{q^2}$:*

1. $F(x, y) = (x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1})$ with $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2$;
2. $F(X) = X^{Q+1}$;
3. $F(X) = X^{Q+q} + \varepsilon X^{Q+1}$ with $\varepsilon \in \mathbb{F}_q \setminus \{1\}$.

Then $r \nmid [C'_F : Z]$.

Proof In the following we claim that $C'_F = C_F$ under the given conditions. Then the desired statement simply follows from Lemma 19.

The inclusion $C_F \subseteq C'_F$ is trivial.

Now let $\lambda = \begin{pmatrix} M & K \\ N & L \end{pmatrix} \in C'_F$. Then (31) holds for $b = a$. Applying similar argument as in the proof of Theorem 22, we note that as long as F is not a permutation, then $K = N = 0$, so $\lambda \in \text{ELM}$ and so $\lambda \in C_F$.

If F is a permutation, the proof of Theorem 22 implies that either $K = N = 0$ or $M = L = 0$. Assume it happens to be the latter case. Then we have $F \circ K \circ F = N \in \text{GL}(\mathbb{F}_{q^2})$. Note that in this case $\frac{k}{\gcd(k, \ell)}$ is odd, and $F(X) = X^{Q+1}$ or X^{Q+q} according to whether $\frac{\ell}{\gcd(k, \ell)}$ is even or odd. However through expanding $F \circ K \circ F$ we find that there are no linear

terms in either case at all, so that $N = 0$ which is a contradiction. Hence we must also have $K = N = 0$, thereby $\lambda \in \text{ELM}$ and $\lambda \in C_F$.

□

Theorem 25 Assume $k > 2, k \notin \{4, 6, 10, 12, 18\}$ and $(Q + 1)^k \not\equiv 0, 1 \pmod{R}$. Let θ be a fixed primitive element of $\mathbb{F}_q^*$. The CCZ-equivalence class representatives of the collection of functions

$$\mathcal{P} := \{(x^{Q+1} + x^Q y, xy^Q + \varepsilon y^{Q+1}) : \varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_2\} \cup \{X^{Q+q} + \varepsilon X^{Q+1} : \varepsilon \in \mathbb{F}_q \setminus \{1\}\} \cup \{X^{Q+1}\}$$

are given by:

1. X^{Q+1} ;
2. X^{Q+q} ;
3. $(x^{Q+1} + x^Q y, xy^Q + \theta^i y^{Q+1})$, where i varies over $\mathcal{C}$;
4. $X^{Q+q} + \theta^i X^{Q+1}$, where i varies over $\mathcal{C}$.

Proof We note that there is a permutation in $\mathcal{P}$ if and only if $\frac{k}{\gcd(k, \ell)}$ is odd, and the only permutation under this situation is either X^{Q+1} or X^{Q+q} according to whether $\frac{\ell}{\gcd(k, \ell)}$ is even or odd. Hence by Lemma 24, Theorem 22 and Remark 4, $f_1, f_2 \in \mathcal{P}$ are CCZ-equivalent if and only if they are linear equivalent via a composition of $\mathcal{L}$ -equivalence and $(\cdot)^{(2^t)}$ for some non-negative integer t . Then the desired result follows from Theorem 15.

□

Finally, combining Theorems 13 and 25 completes the proof of Statement 2(b) and thereby the whole of Theorem 2.

Remark 9 We have checked the exact CCZ-equivalence of the butterflies for $k = 3, 5$ by Magma experiment. When $k \geq 7$, the program runs out of time. In such a case, for given $(\alpha, \beta) \in \mathbb{F}_q^*$ such that $\Theta_1 \neq 0$ and $\Theta_2^Q \neq \Theta_1^{Q-1} \Theta_3$, we may determine precisely which $i \in \mathcal{C}$ such that F_i or f_i in Statement 2 of Theorem 2 $\vee_{\alpha, \beta}^{Q+1}$ is equivalent to by evaluating the ramification and branch points of the rational function $h(x)$ (or $g(X)$) corresponding to, and

then using degree-one rational functions to transfer to one corresponding to a function of the form F_i or f_i in Statement 2 of Theorem 2. However this method is not efficient to make complete classification for general k . Our best knowledge about linear equivalence between two general closed butterflies for any fixed k and ℓ is that $V_{\alpha,\beta}^{Q+1} \sim_L V_{\alpha^2,\beta^2}^{Q+1}$ and when $\alpha \neq 1$, $V_{\alpha,\beta}^{Q+1} \sim_L V_{\alpha,(\alpha+1)^{2Q+2}\beta^{-1}}^{Q+1}$ (see [7] for the proof), but these do not reduce the remaining work required much.

Acknowledgements. The author is grateful to Maosheng Xiong and Lukas Kölsch for valuable suggestions to improve and enhance the results, and would like to thank the Department of Mathematics, Hong Kong University of Science and Technology for financial support. The main research result and the start of manuscript drafting were made in the Hong Kong University of Science and Technology, and further proofread and submission of the manuscript are completed in Hetao Institute of Mathematics and Interdisciplinary Sciences.

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